

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

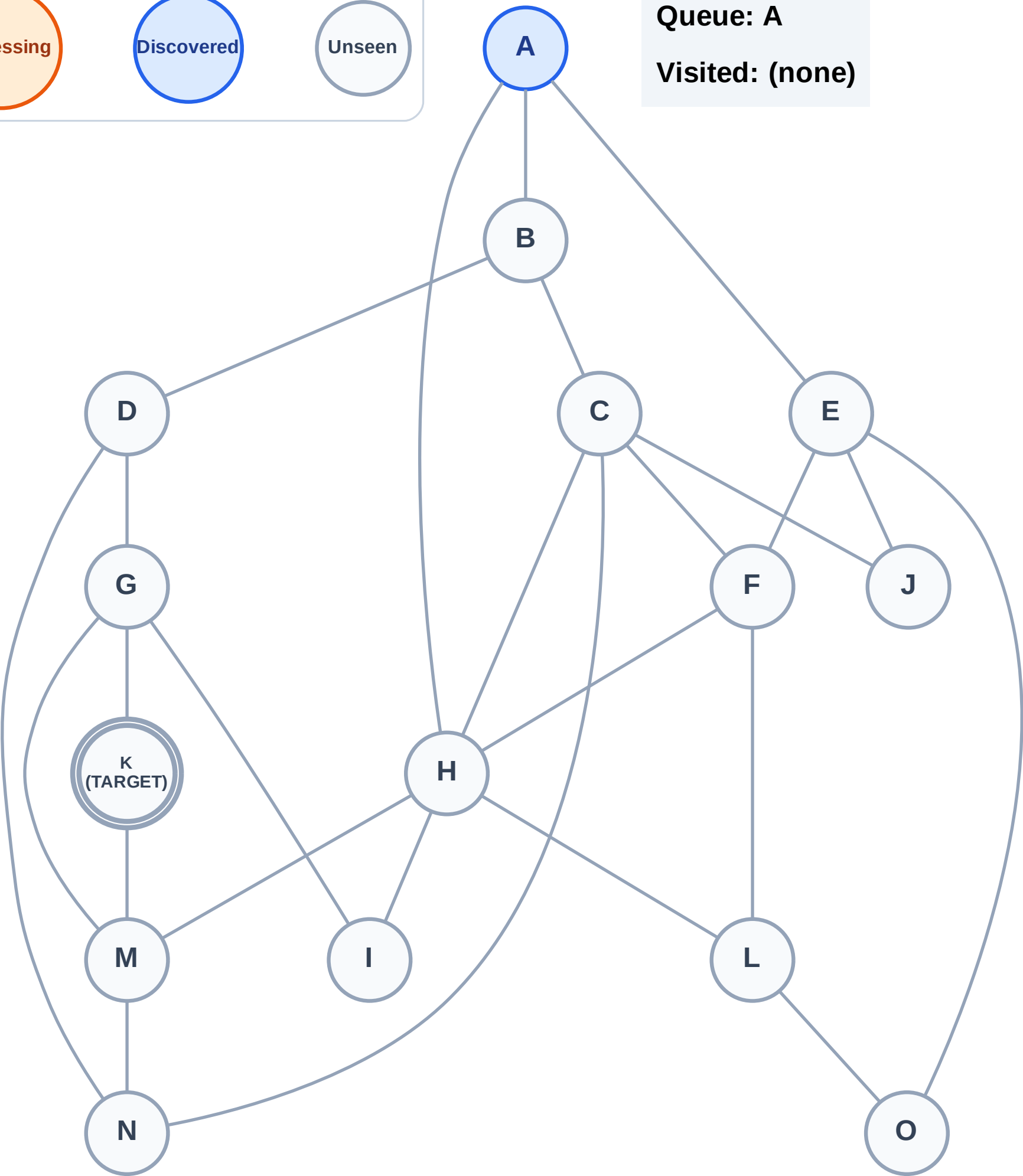
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

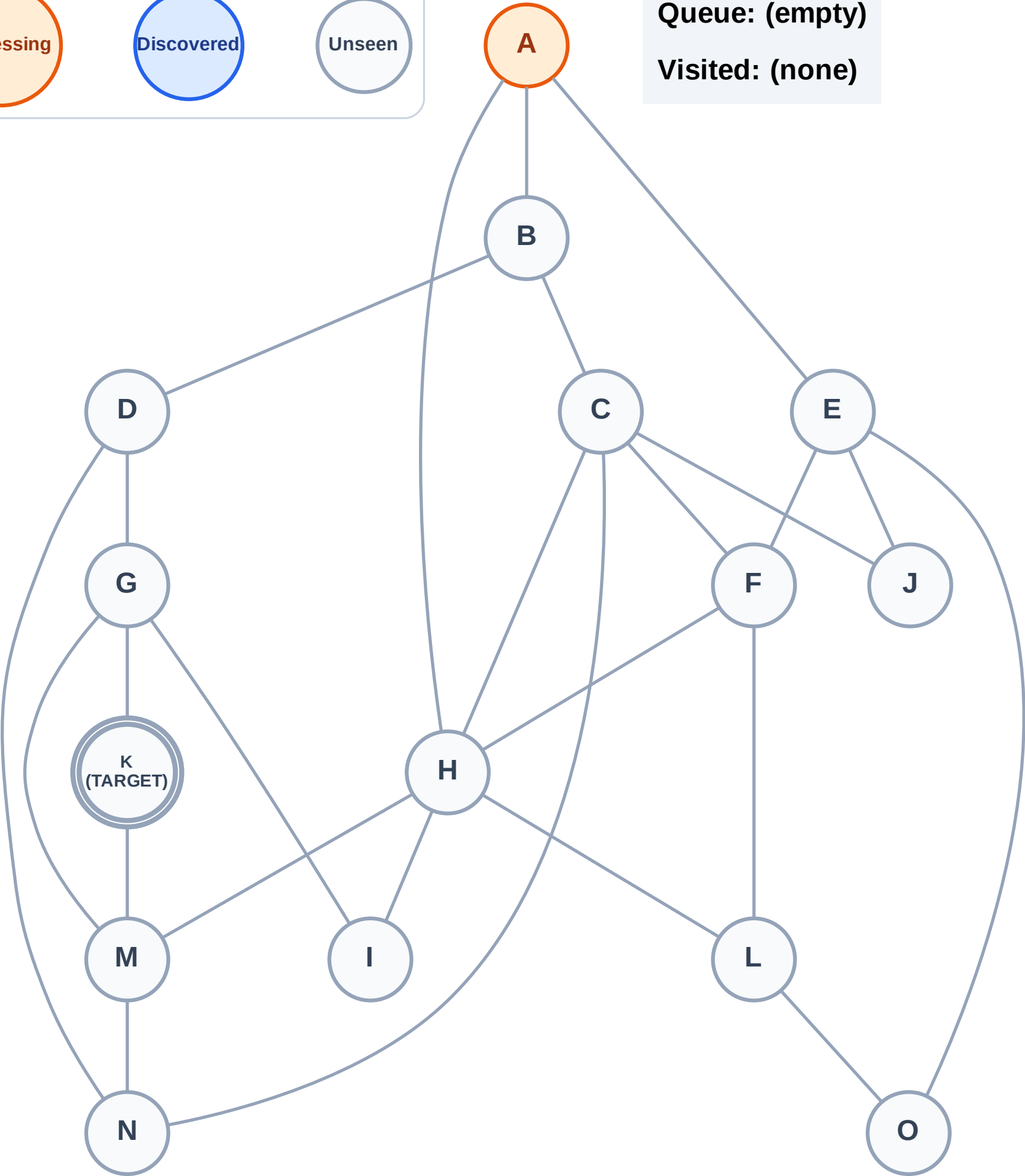
Complete

Processing

Discovered

Unseen

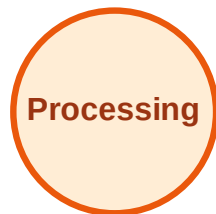
Queue: (empty)
Visited: (none)



BFS Traversal

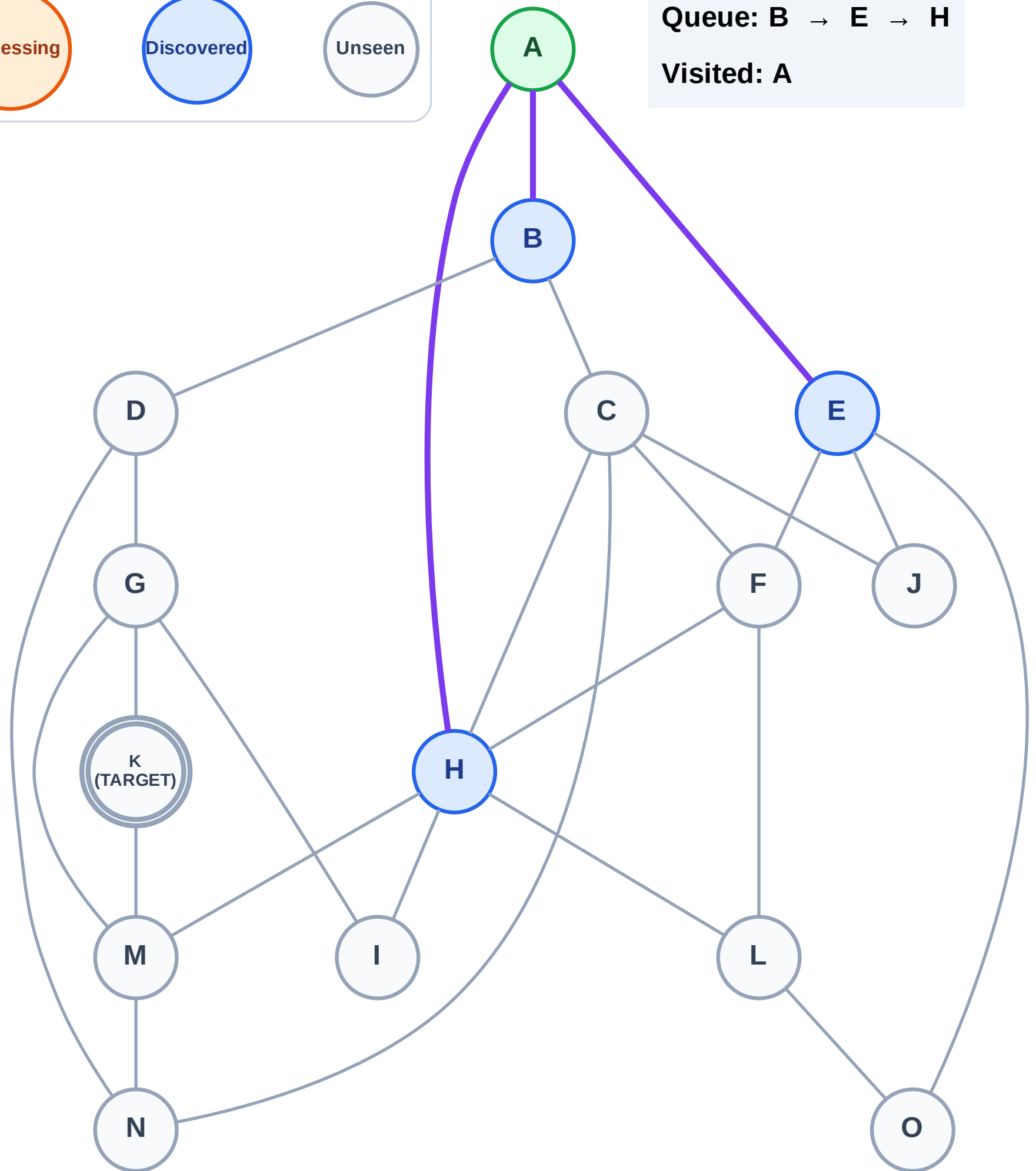
Step 3: Discover B, E, H from A

Legend



Queue: B → E → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

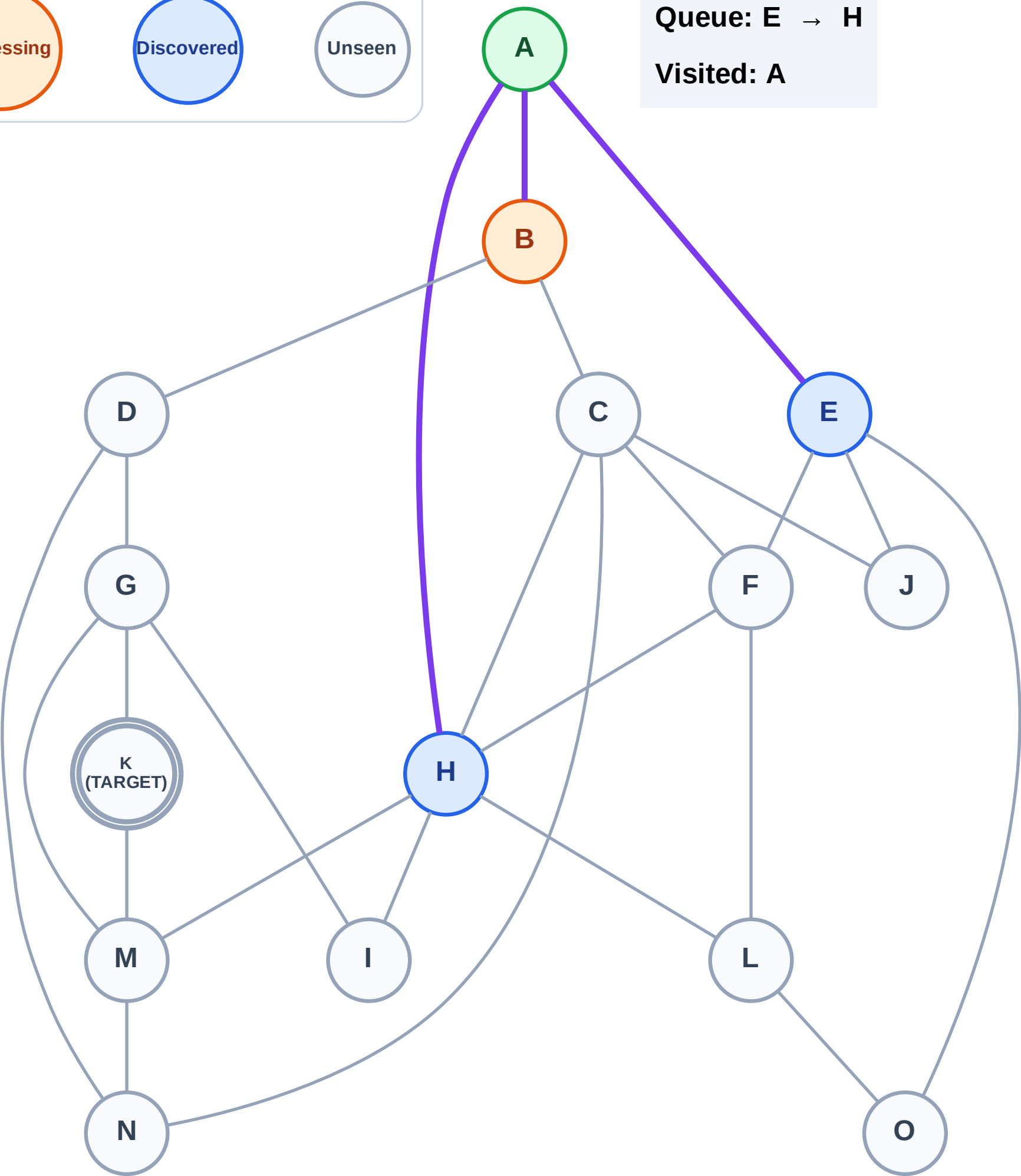
Processing

Discovered

Unseen

Queue: E → H

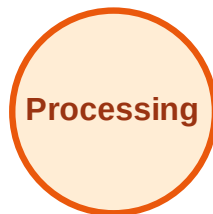
Visited: A



BFS Traversal

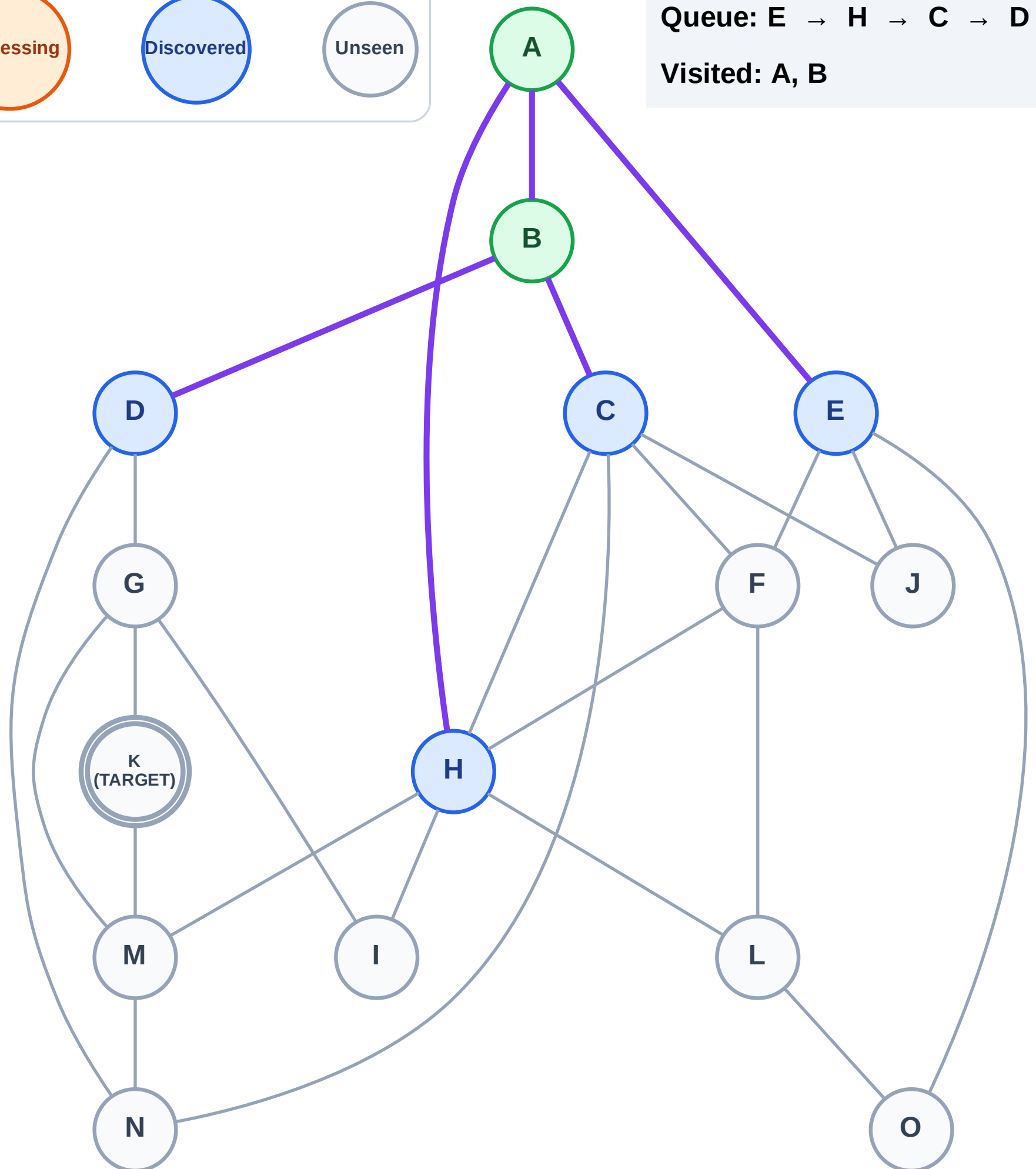
Step 5: Discover C, D from B

Legend



Queue: E → H → C → D

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

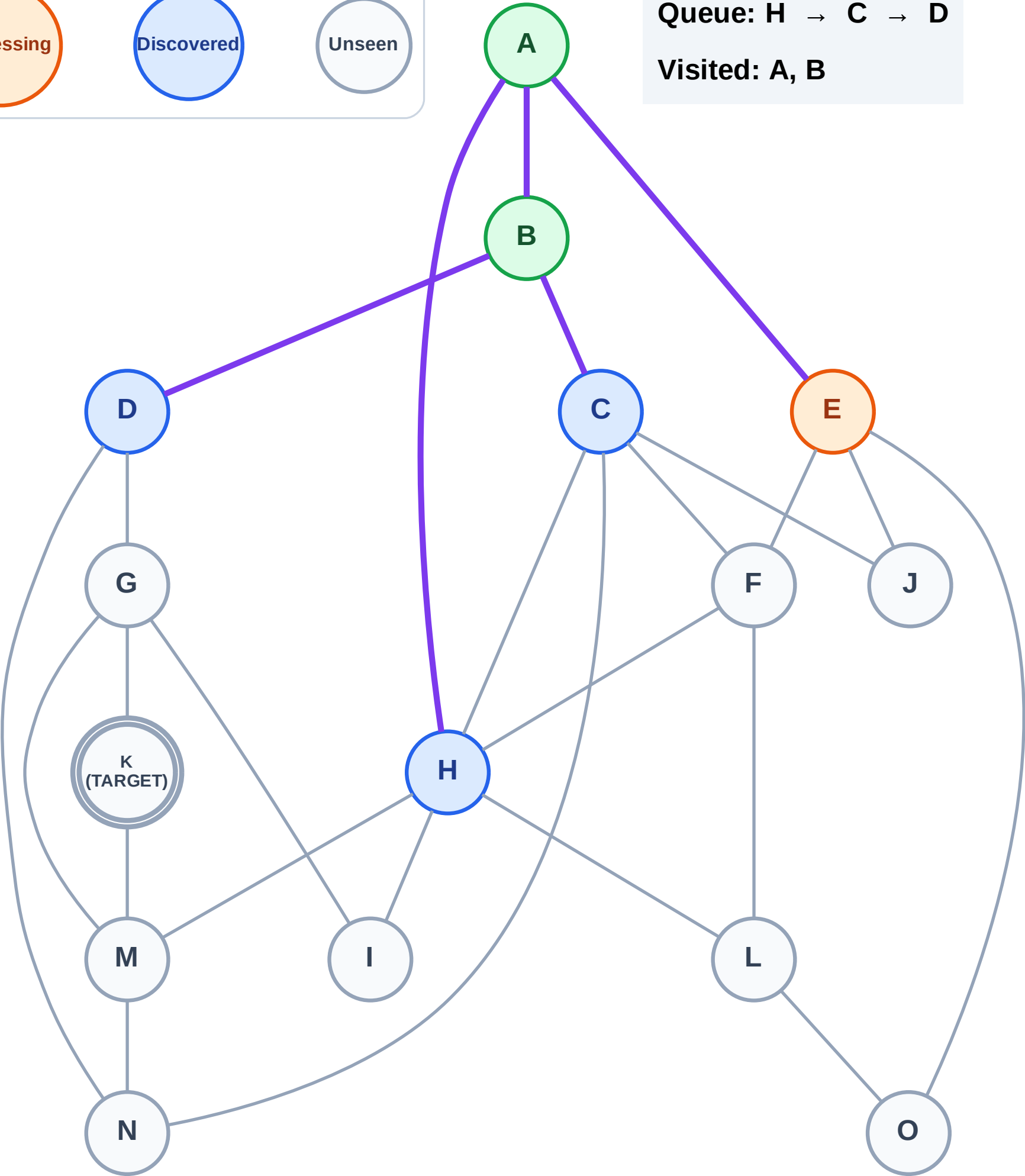
Processing

Discovered

Unseen

Queue: H → C → D

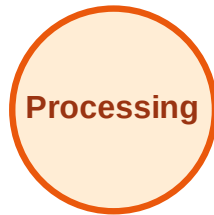
Visited: A, B



BFS Traversal

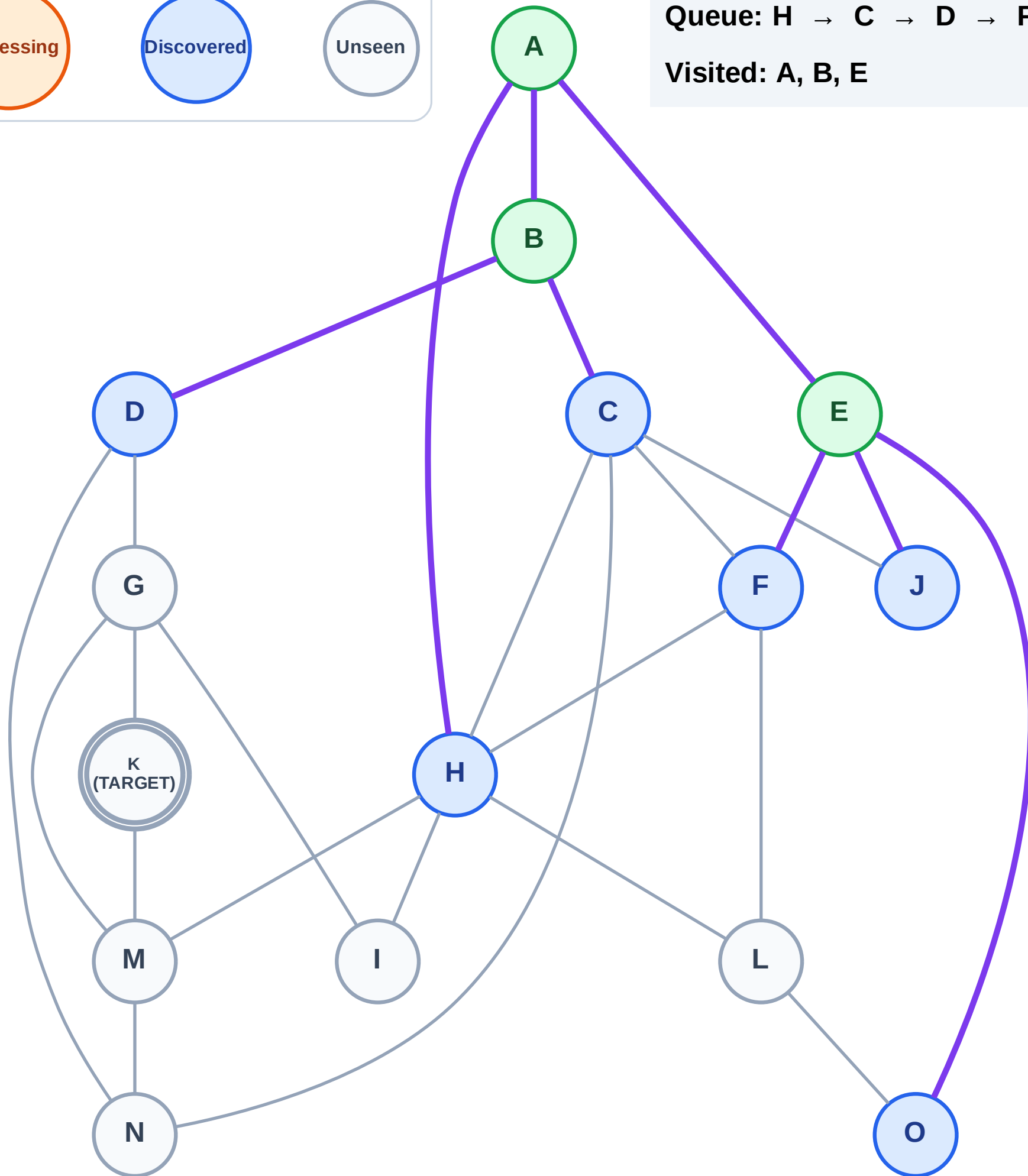
Step 7: Discover F, J, O from E

Legend



Queue: H → C → D → F → J → O

Visited: A, B, E



BFS Traversal

Step 8: Process node H

Legend



Complete



Processing



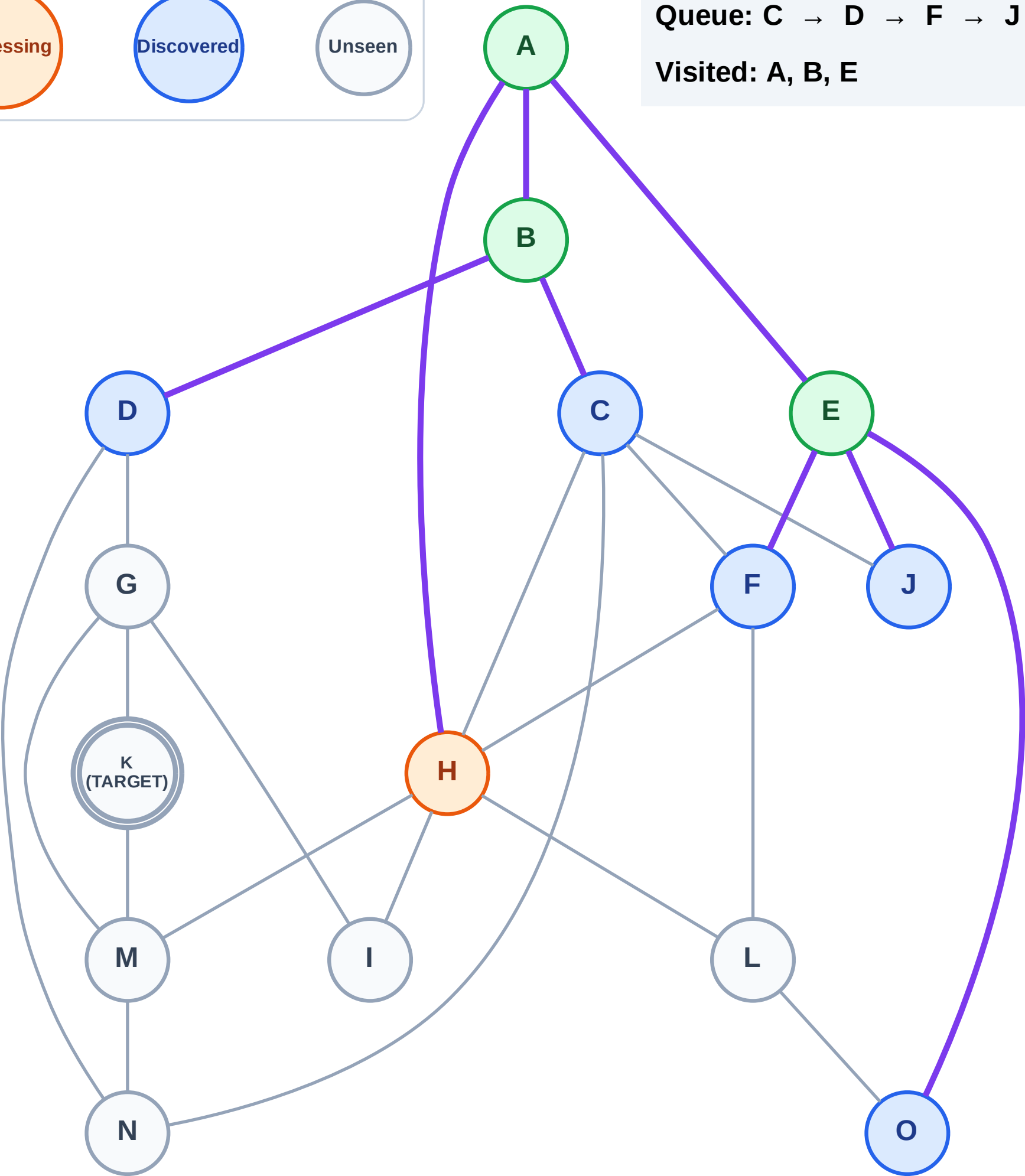
Discovered



Unseen

Queue: C → D → F → J → O

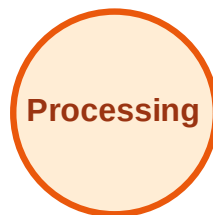
Visited: A, B, E



BFS Traversal

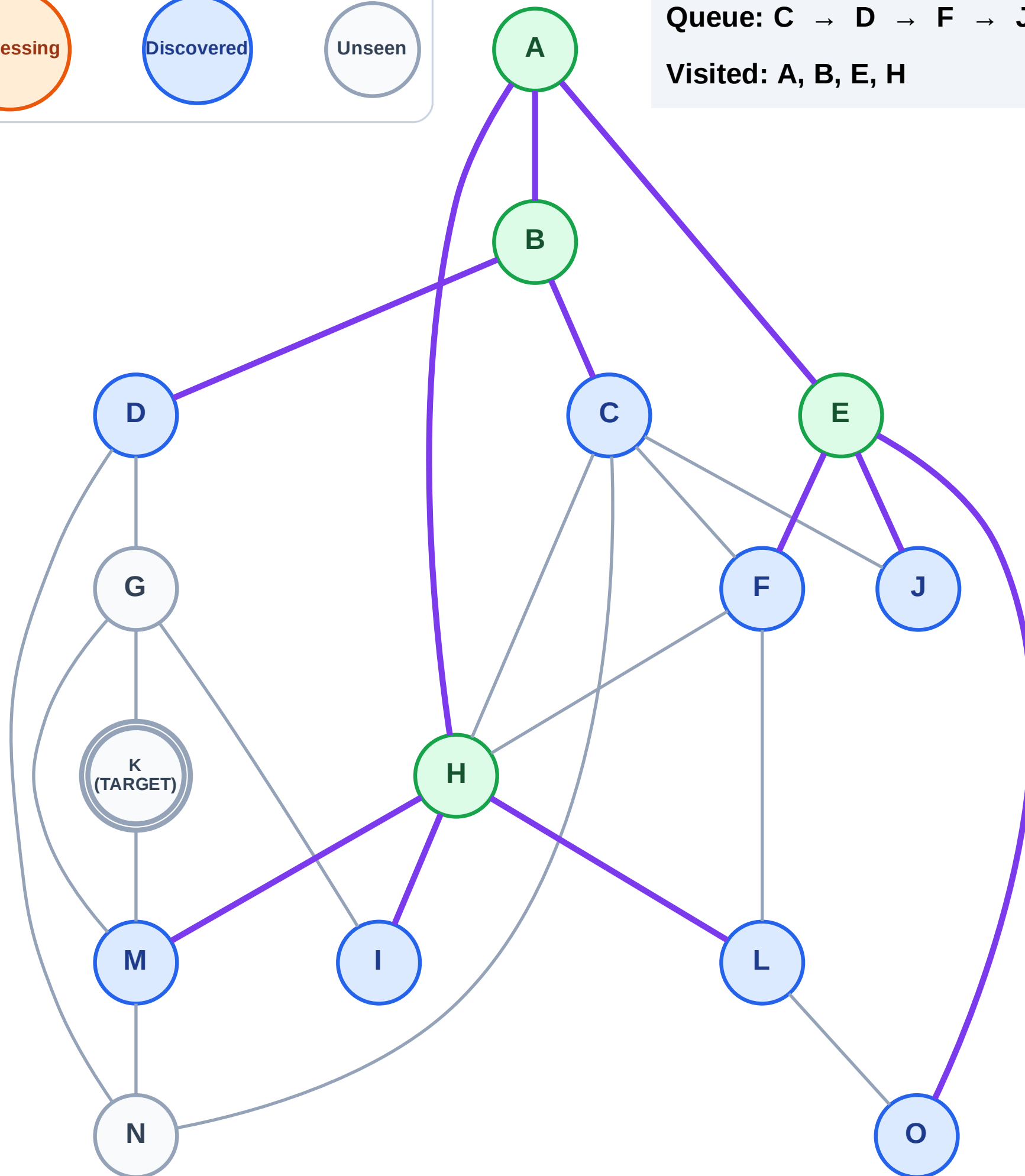
Step 9: Discover I, L, M from H

Legend

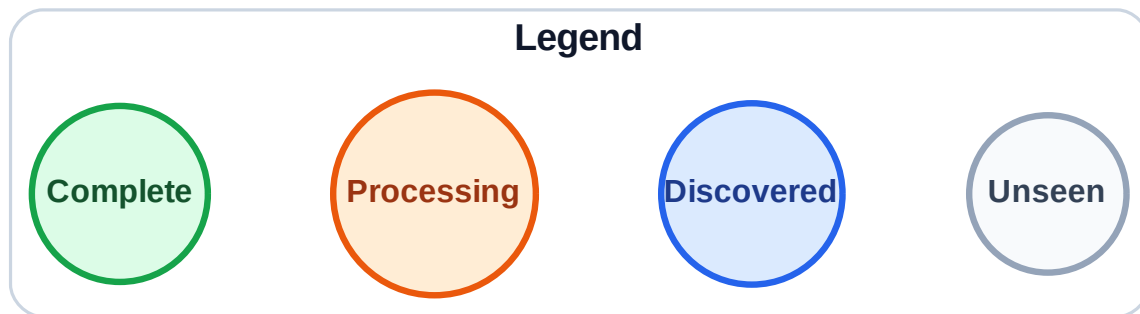


Queue: C → D → F → J → O → I → L → M

Visited: A, B, E, H

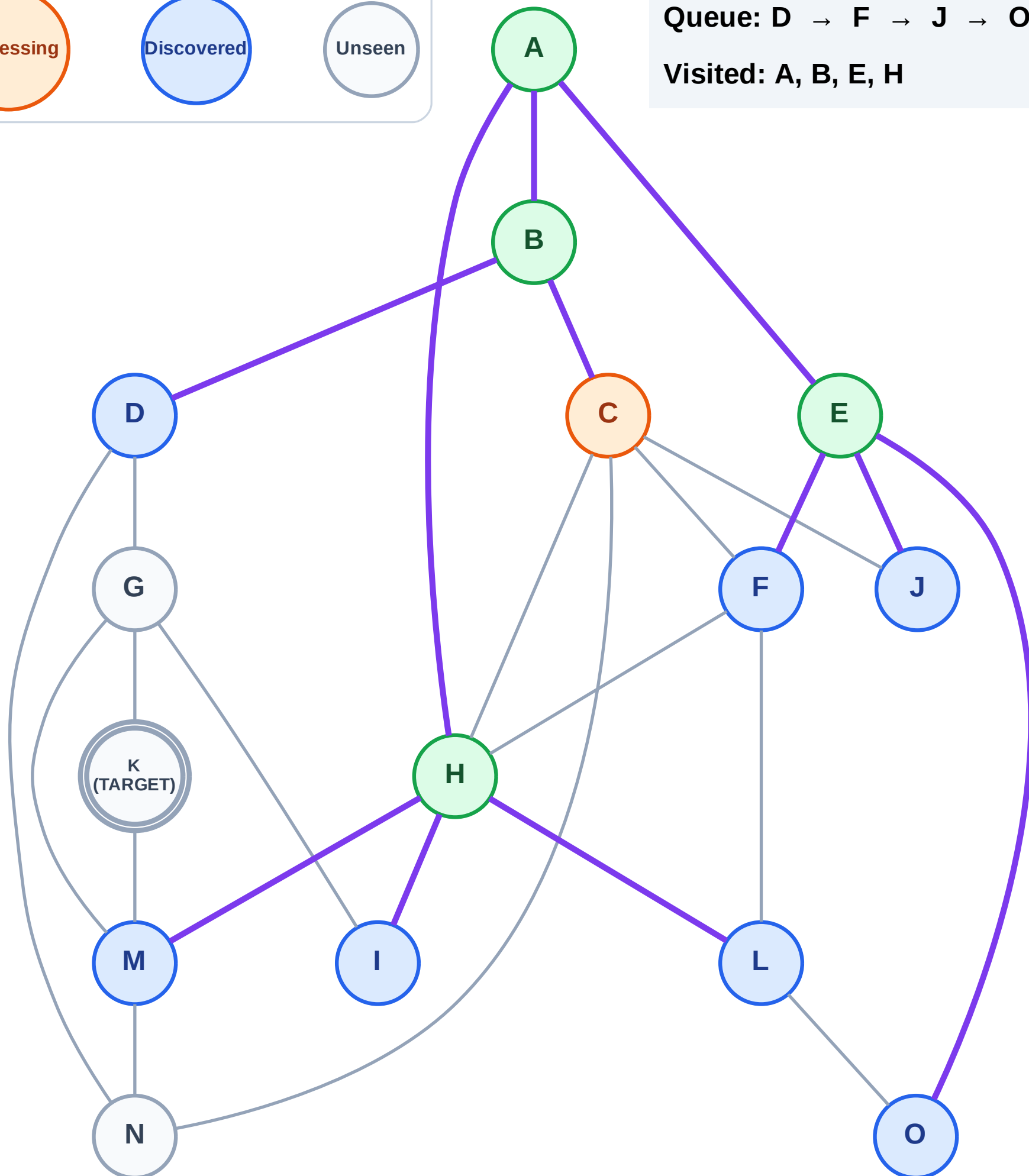


Step 10: Process node C



Queue: D → F → J → O → I → L → M

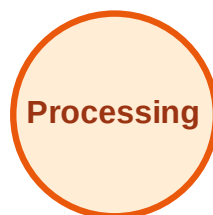
Visited: A, B, E, H



BFS Traversal

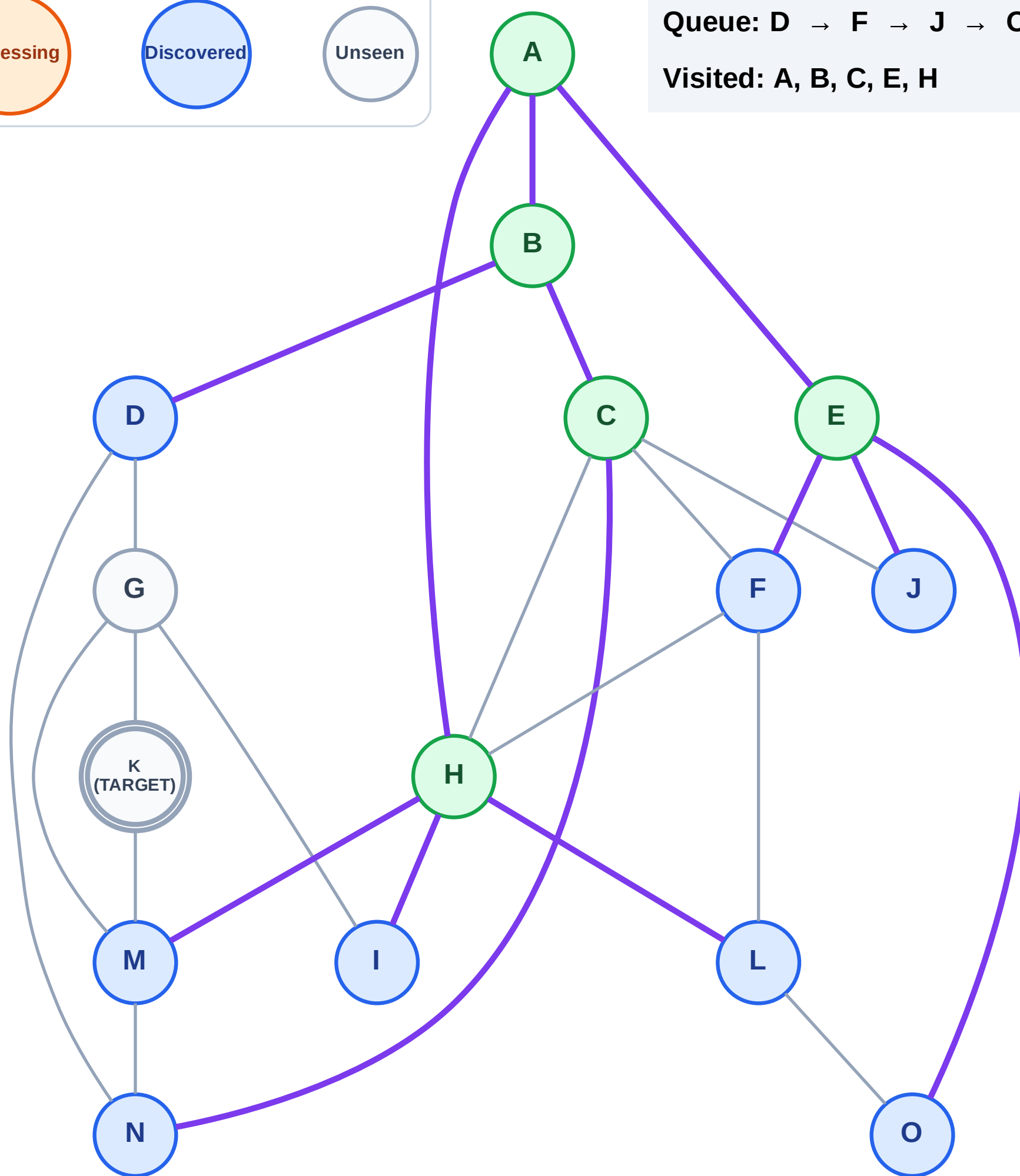
Step 11: Discover N from C

Legend

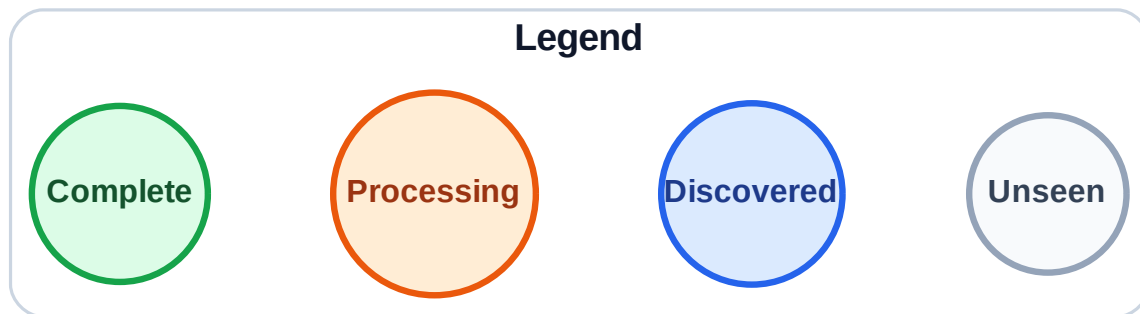


Queue: D → F → J → O → I → L → M → N

Visited: A, B, C, E, H

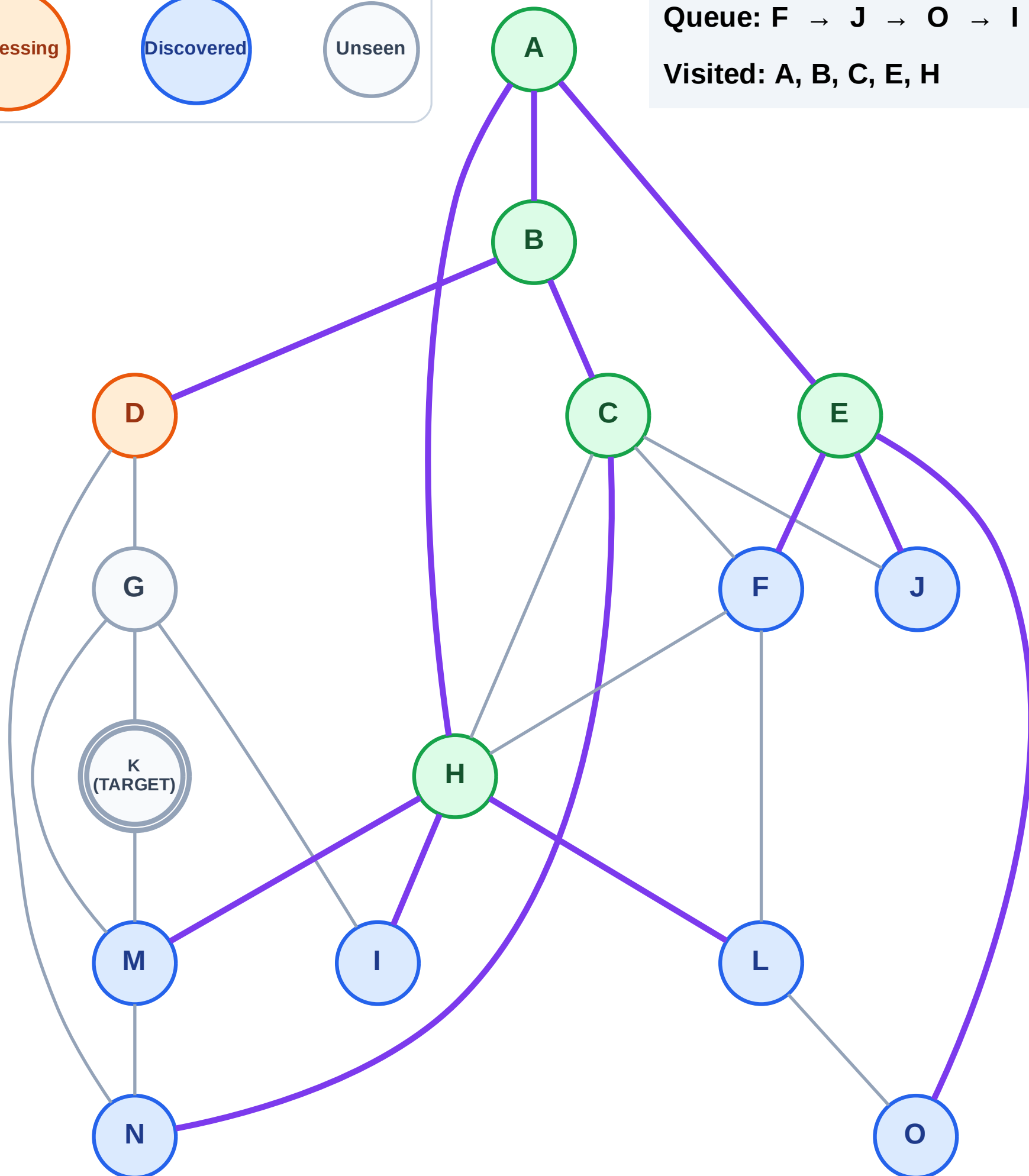


Step 12: Process node D



Queue: F → J → O → I → L → M → N

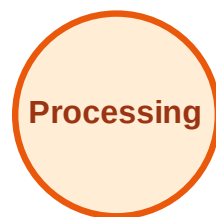
Visited: A, B, C, E, H



BFS Traversal

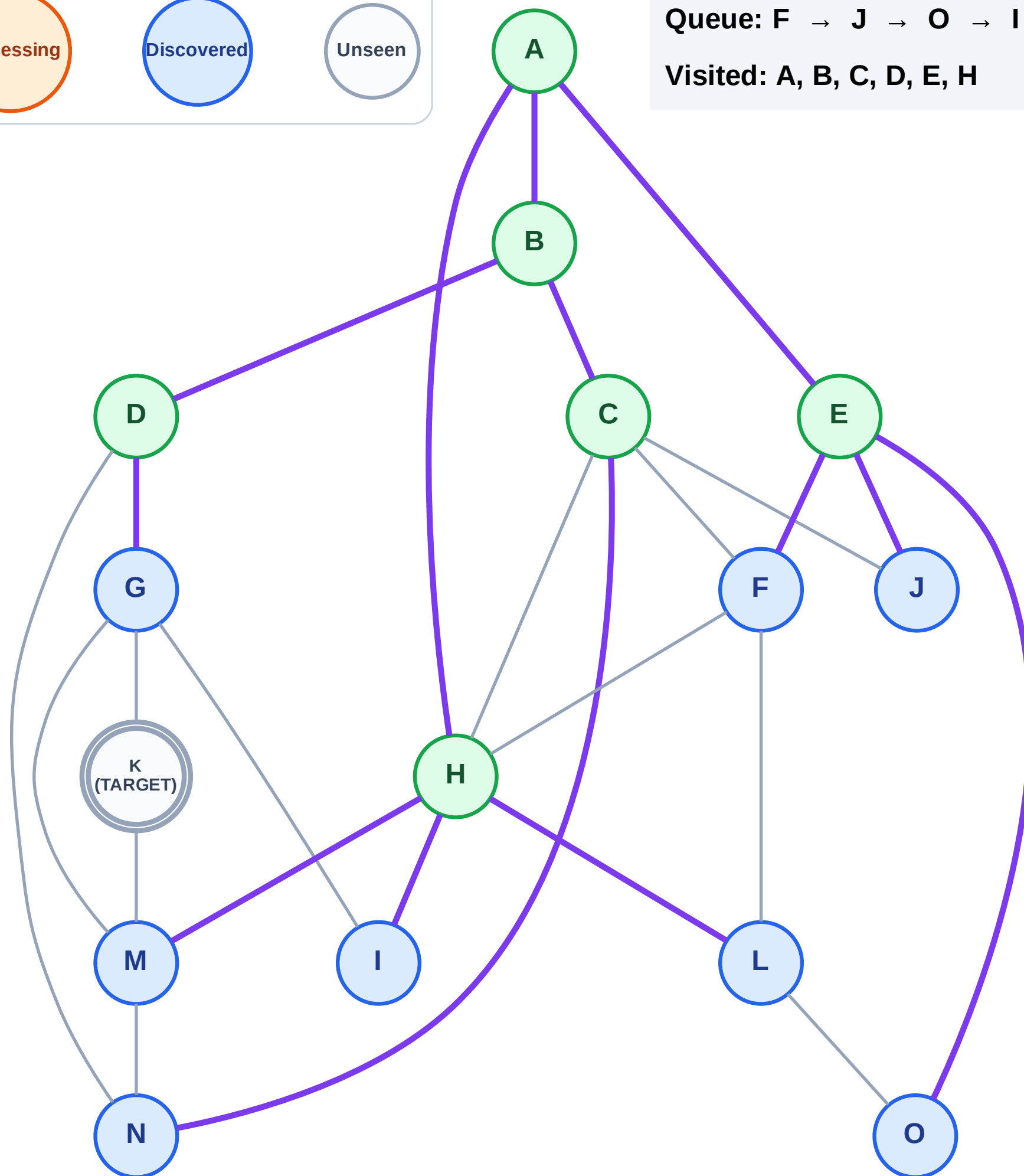
Step 13: Discover G from D

Legend



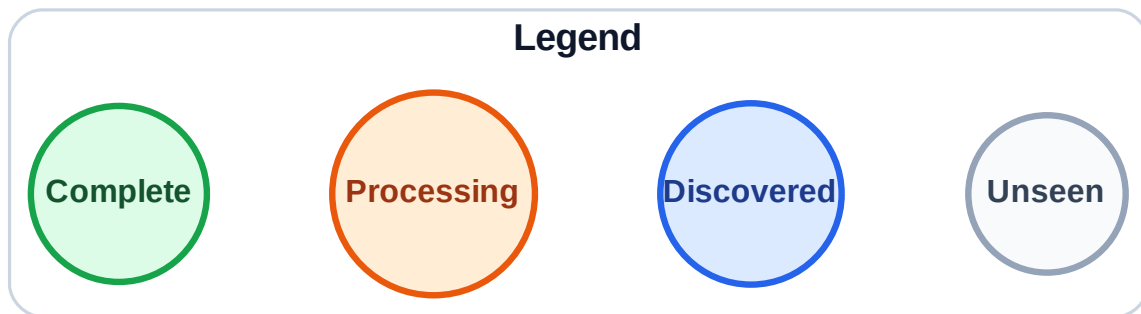
Queue: F → J → O → I → L → M → N → G

Visited: A, B, C, D, E, H



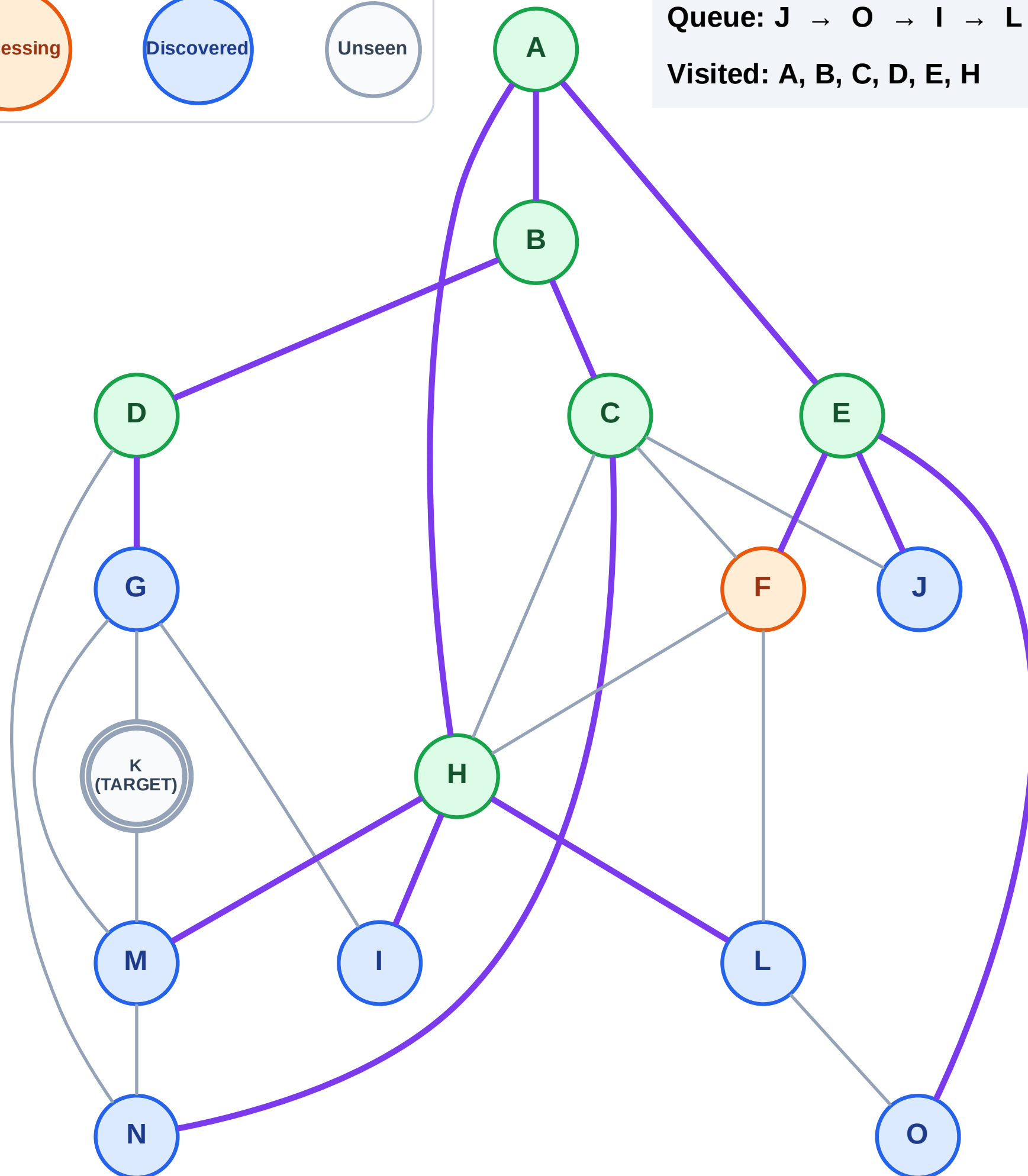
Step 14: Process node F

Step 14: Process node F



Queue: J → O → I → L → M → N → G

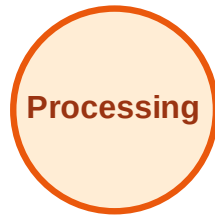
Visited: A, B, C, D, E, H



BFS Traversal

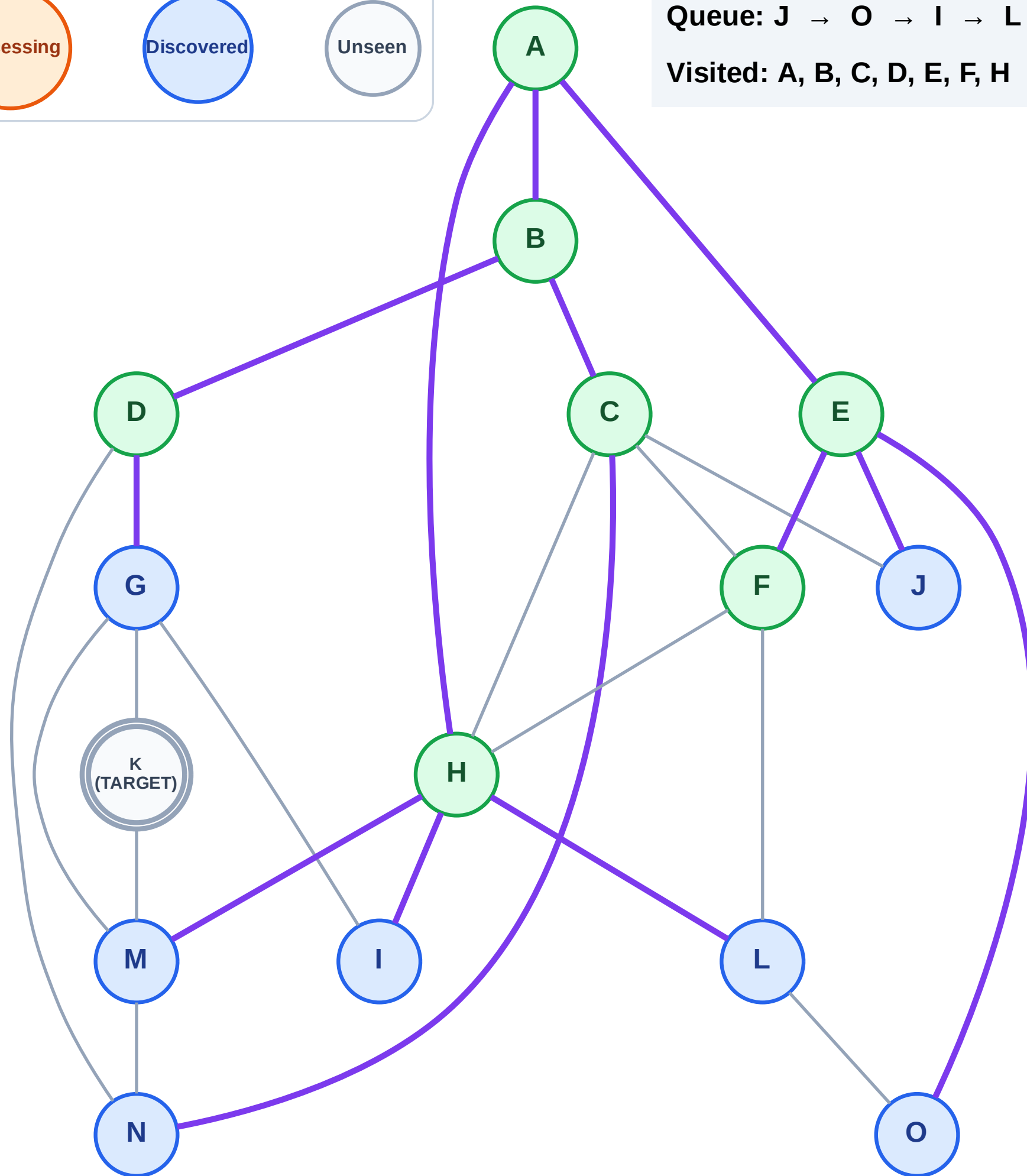
Step 15: No new neighbors from F

Legend



Queue: J → O → I → L → M → N → G

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node J

Legend

Complete

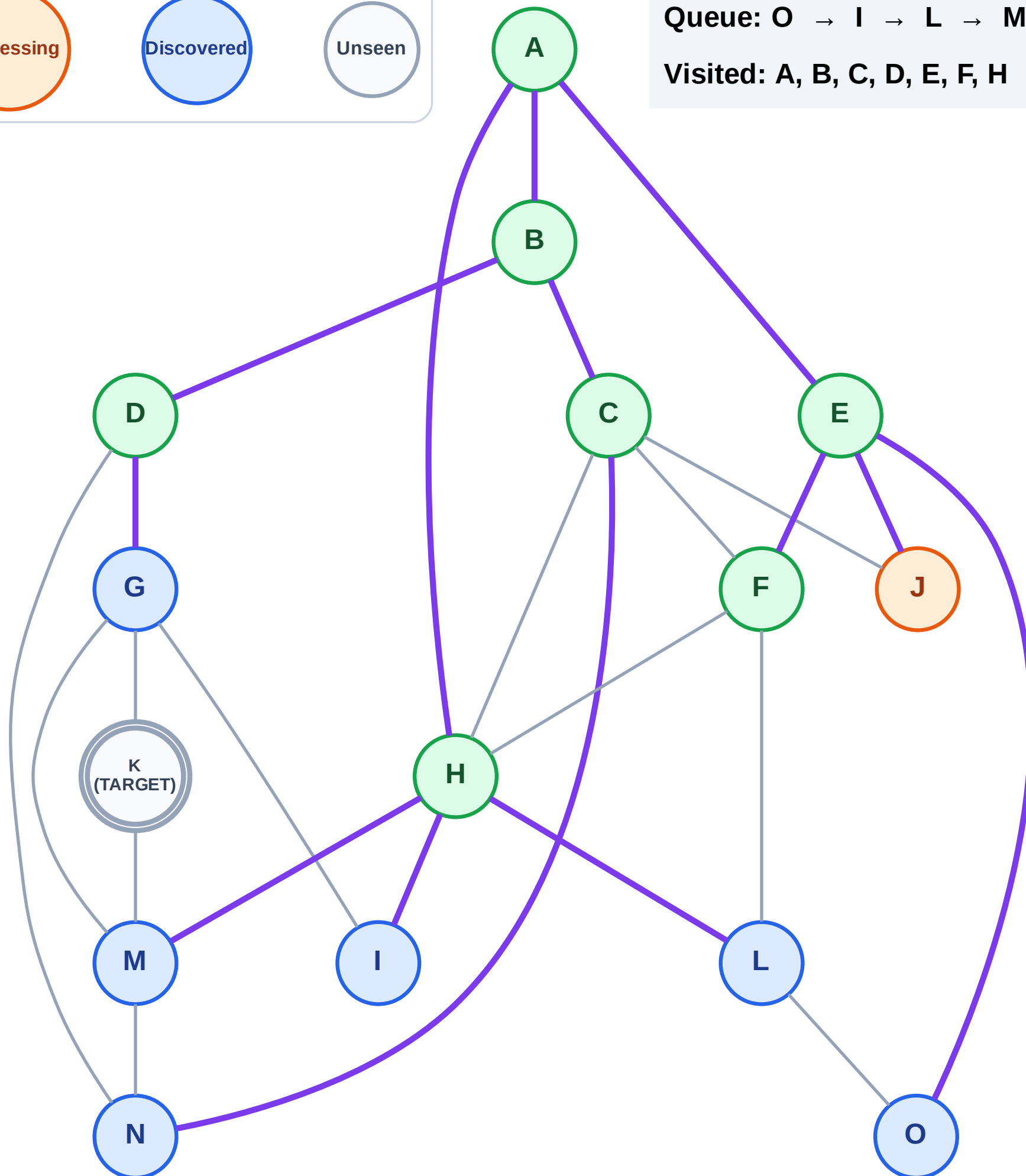
Processing

Discovered

Unseen

Queue: O → I → L → M → N → G

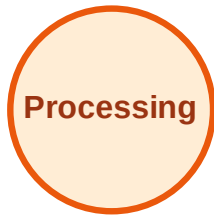
Visited: A, B, C, D, E, F, H



BFS Traversal

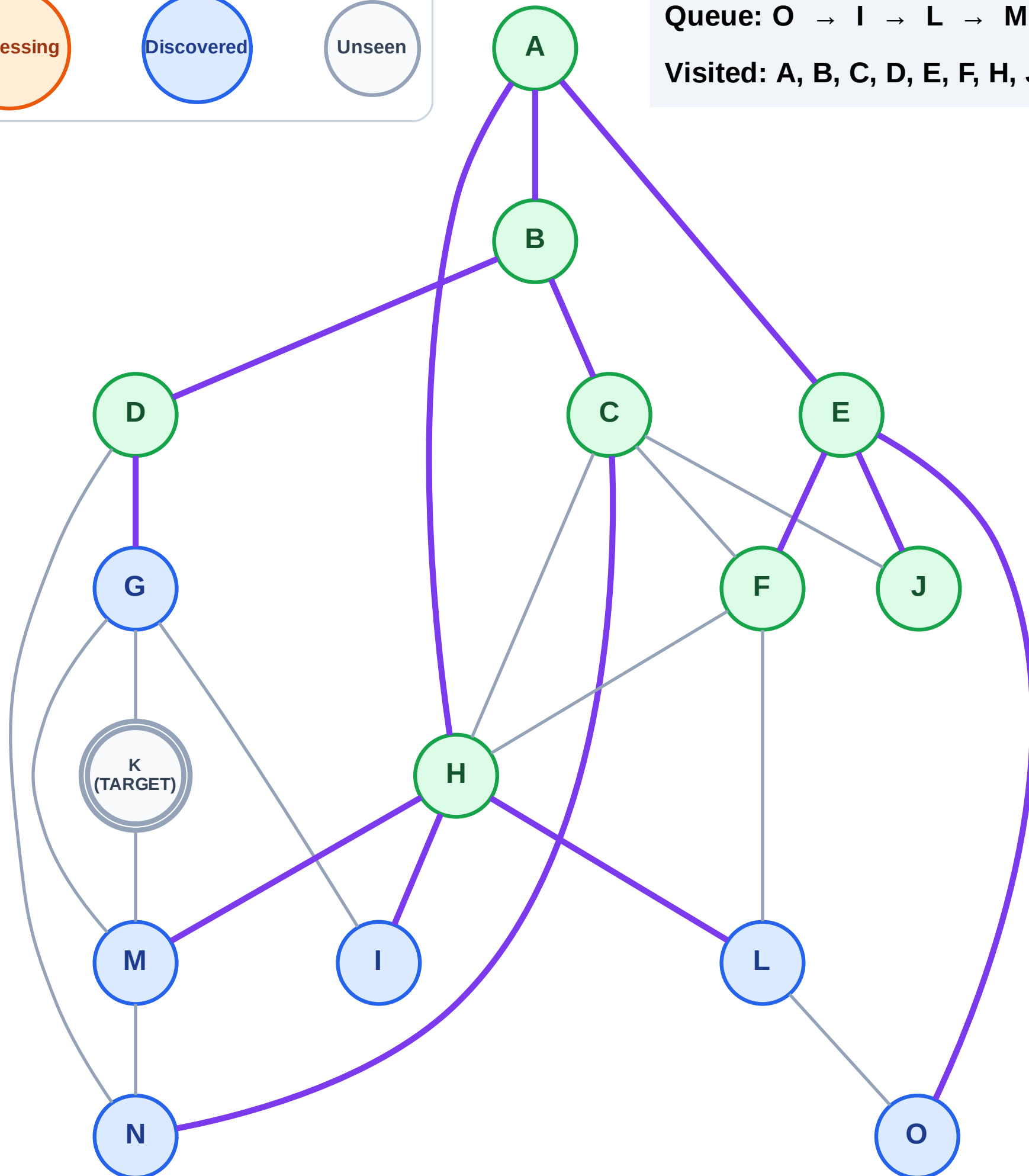
Step 17: No new neighbors from J

Legend



Queue: O → I → L → M → N → G

Visited: A, B, C, D, E, F, H, J



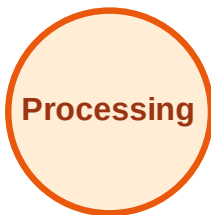
BFS Traversal

Step 18: Process node O

Legend



Complete



Processing



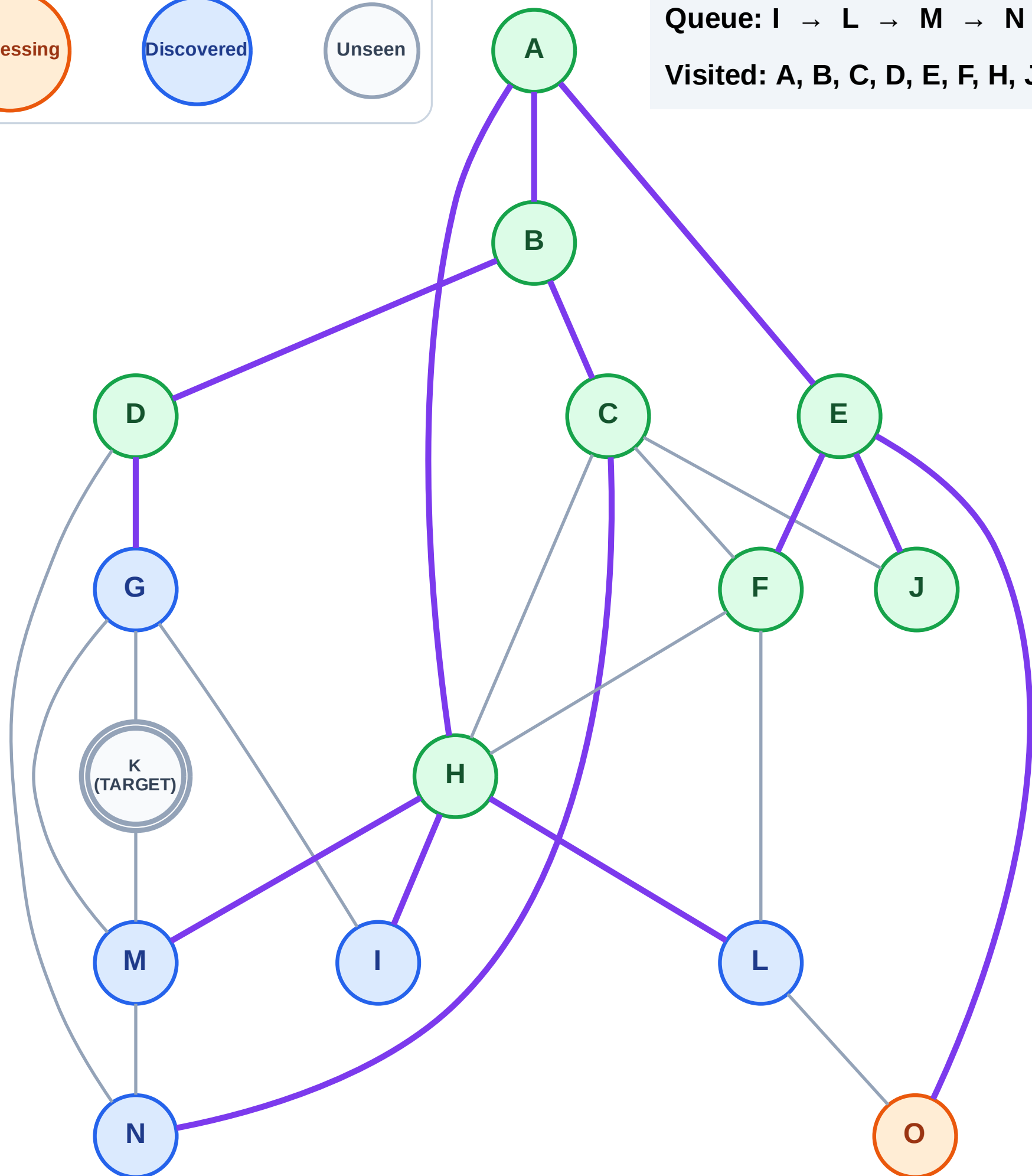
Discovered



Unseen

Queue: I → L → M → N → G

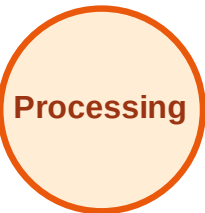
Visited: A, B, C, D, E, F, H, J



BFS Traversal

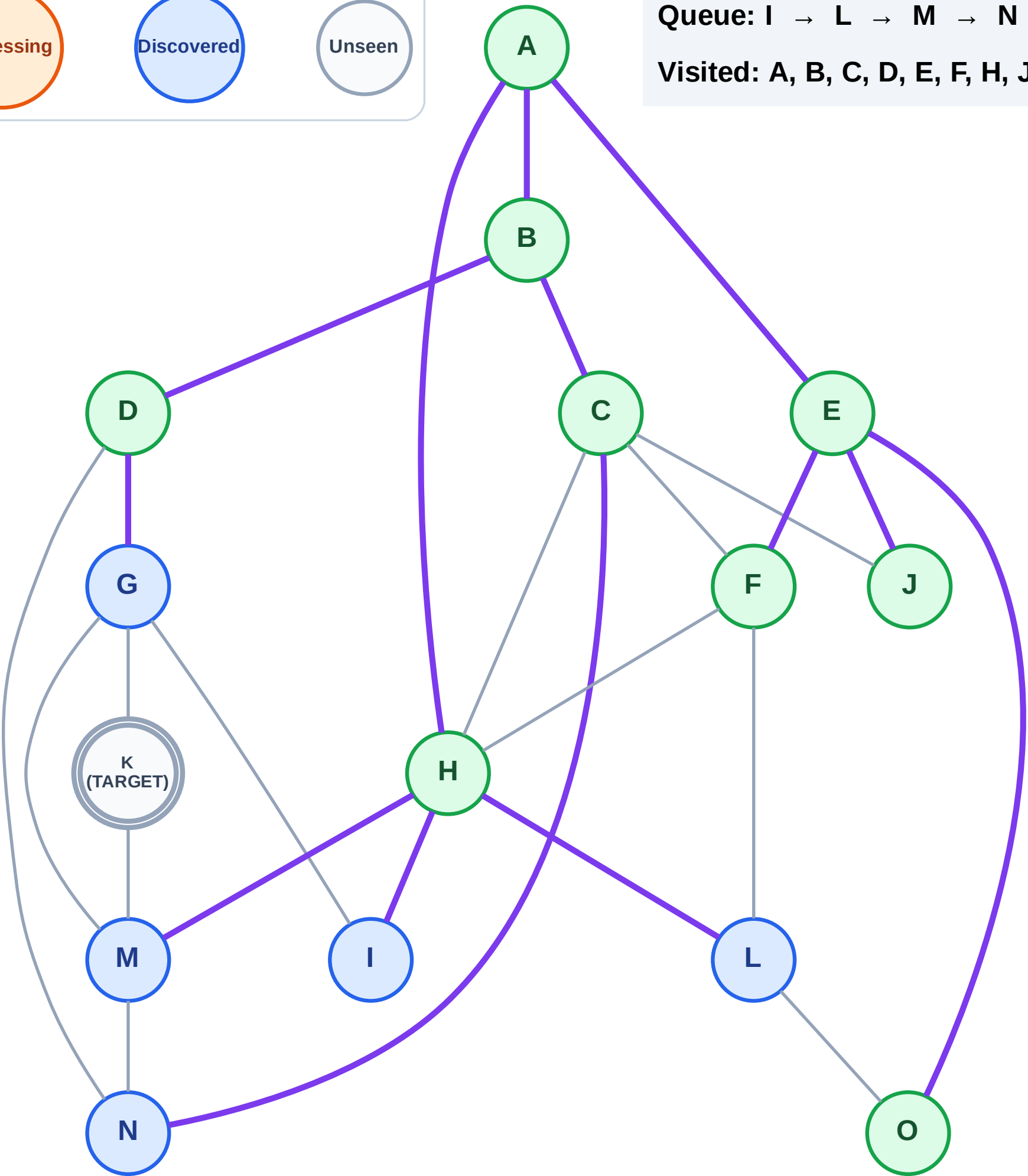
Step 19: No new neighbors from O

Legend



Queue: I → L → M → N → G

Visited: A, B, C, D, E, F, H, J, O



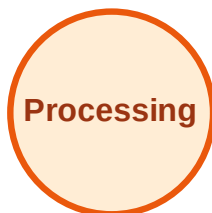
BFS Traversal

Step 20: Process node I

Legend



Complete



Processing



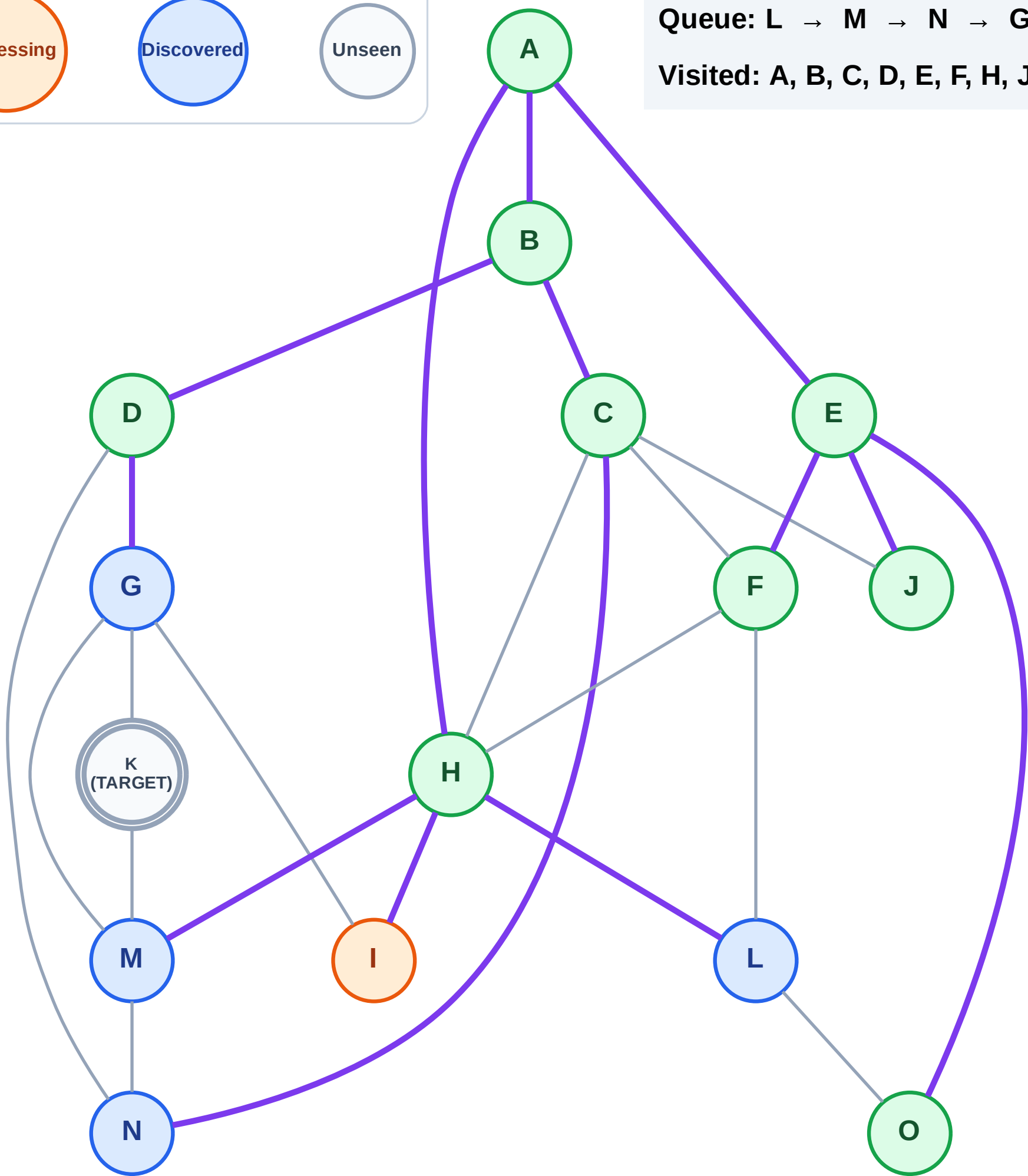
Discovered



Unseen

Queue: L → M → N → G

Visited: A, B, C, D, E, F, H, J, O



BFS Traversal

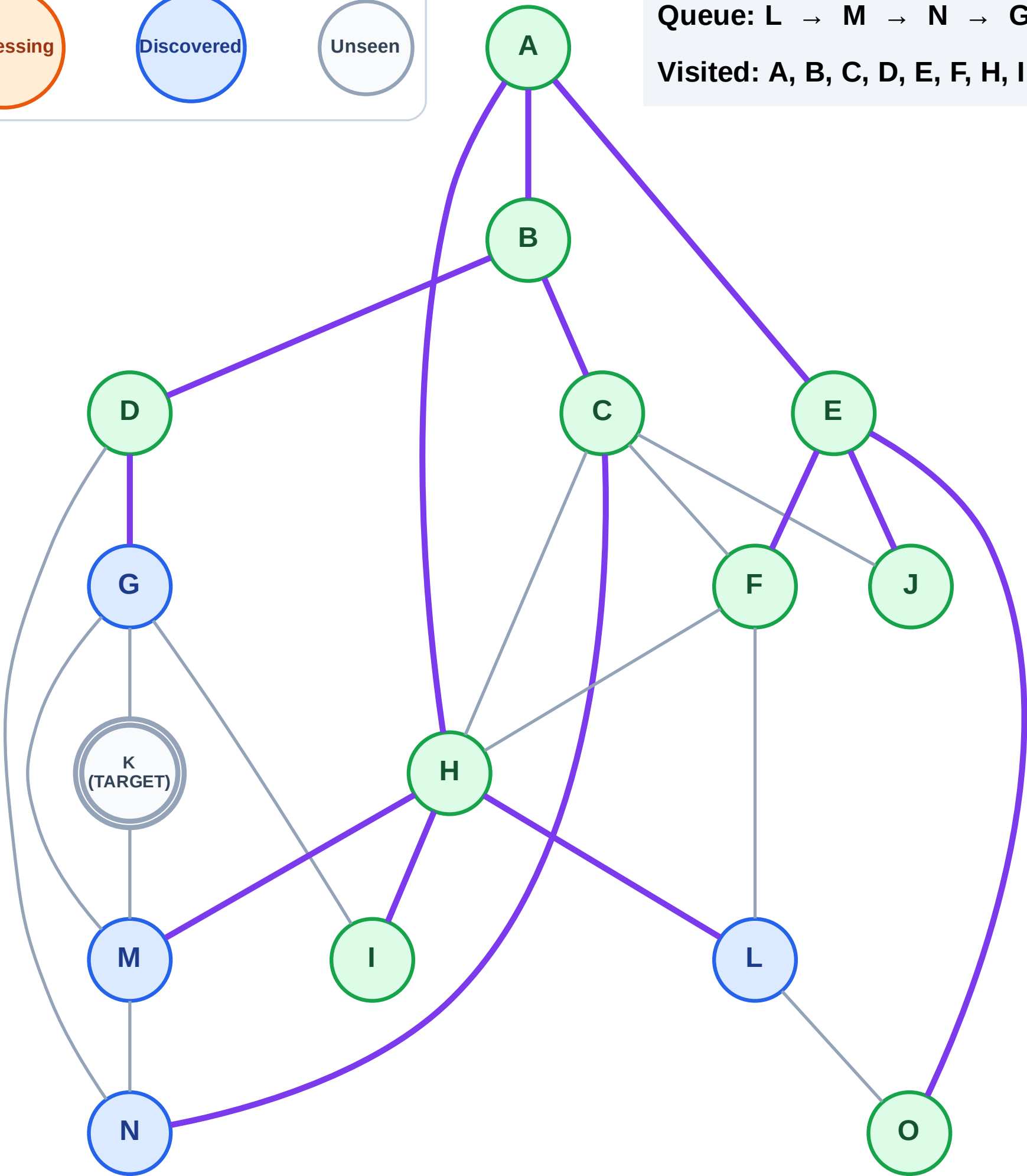
Step 21: No new neighbors from I

Legend



Queue: L → M → N → G

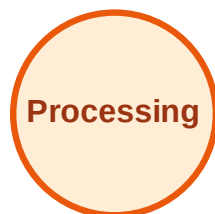
Visited: A, B, C, D, E, F, H, I, J, O



BFS Traversal

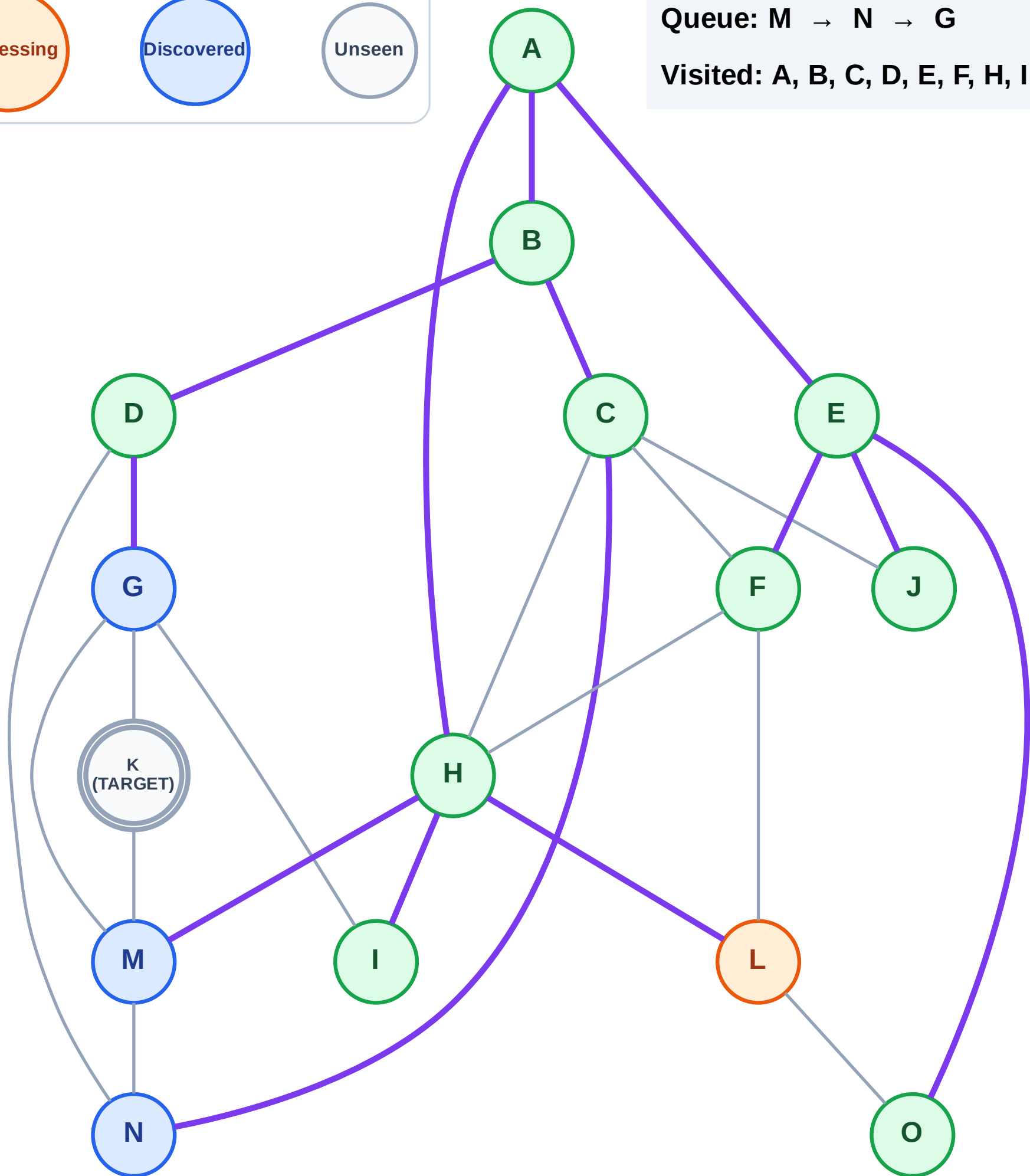
Step 22: Process node L

Legend



Queue: M → N → G

Visited: A, B, C, D, E, F, H, I, J, O



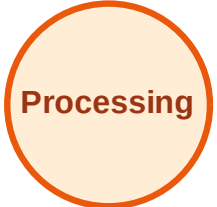
BFS Traversal

Step 23: No new neighbors from L

Legend



Complete



Processing



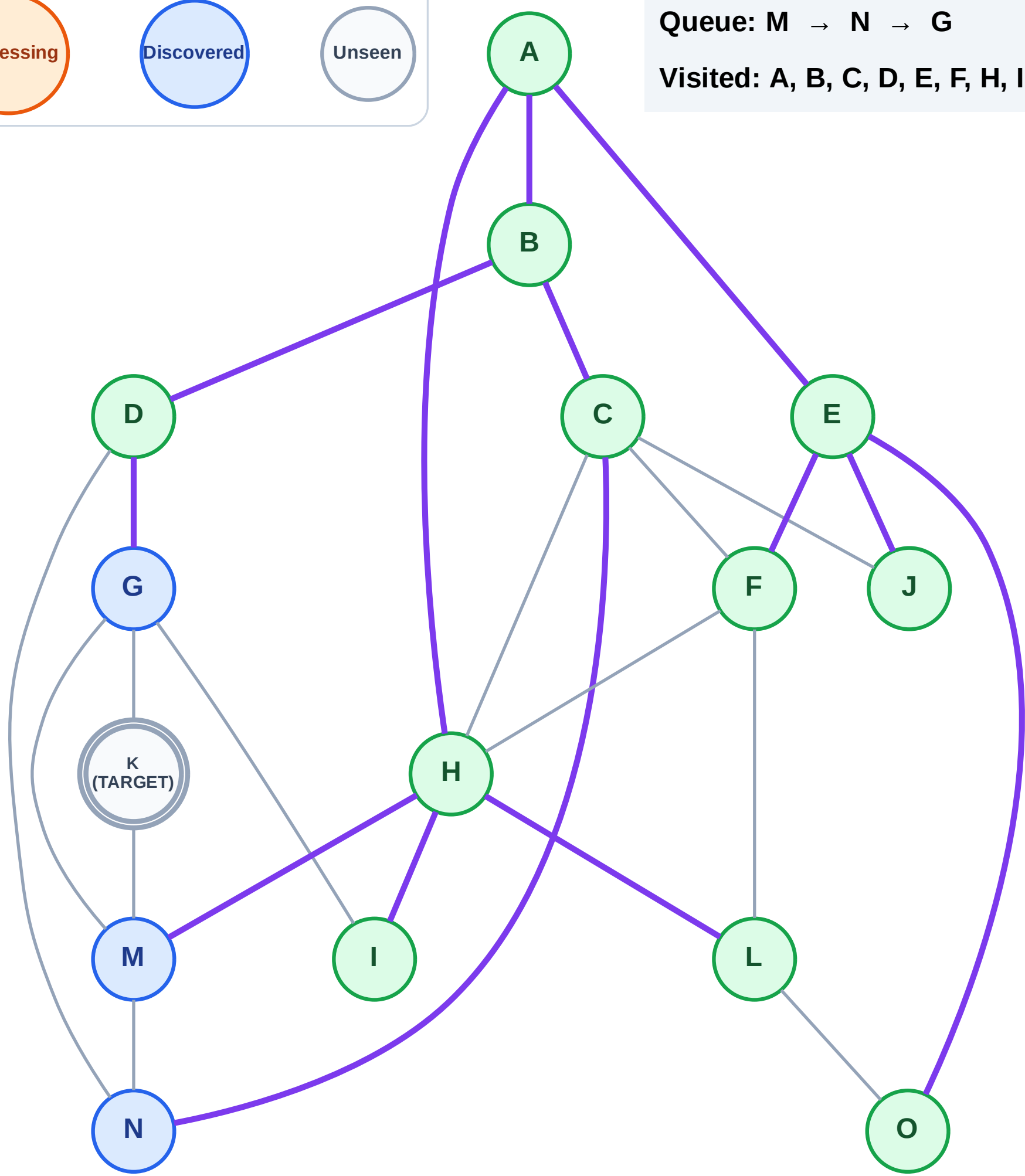
Discovered



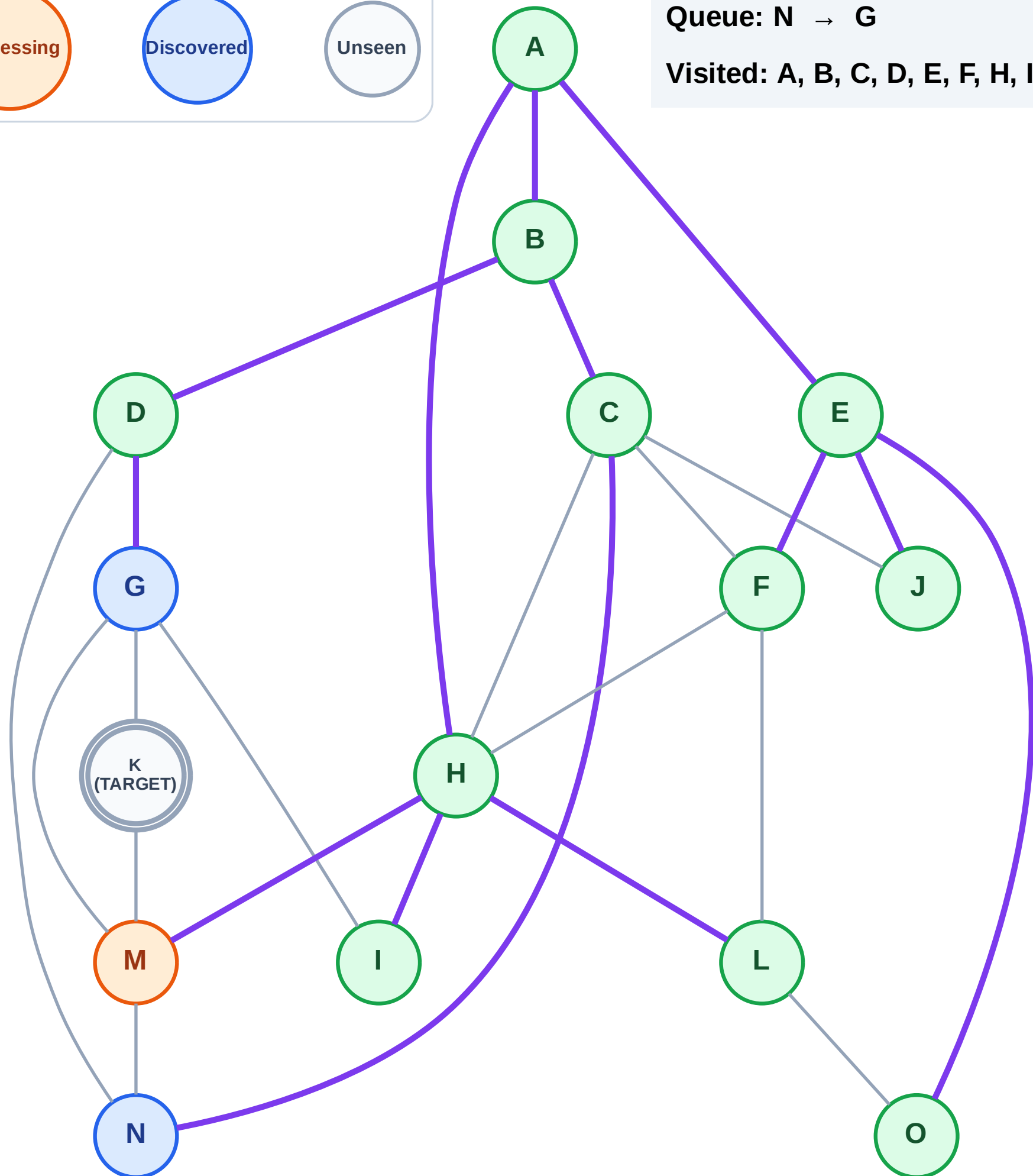
Unseen

Queue: M → N → G

Visited: A, B, C, D, E, F, H, I, J, L, O



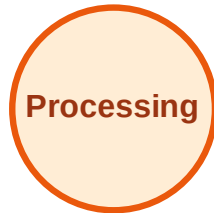
Step 24: Process node M



BFS Traversal

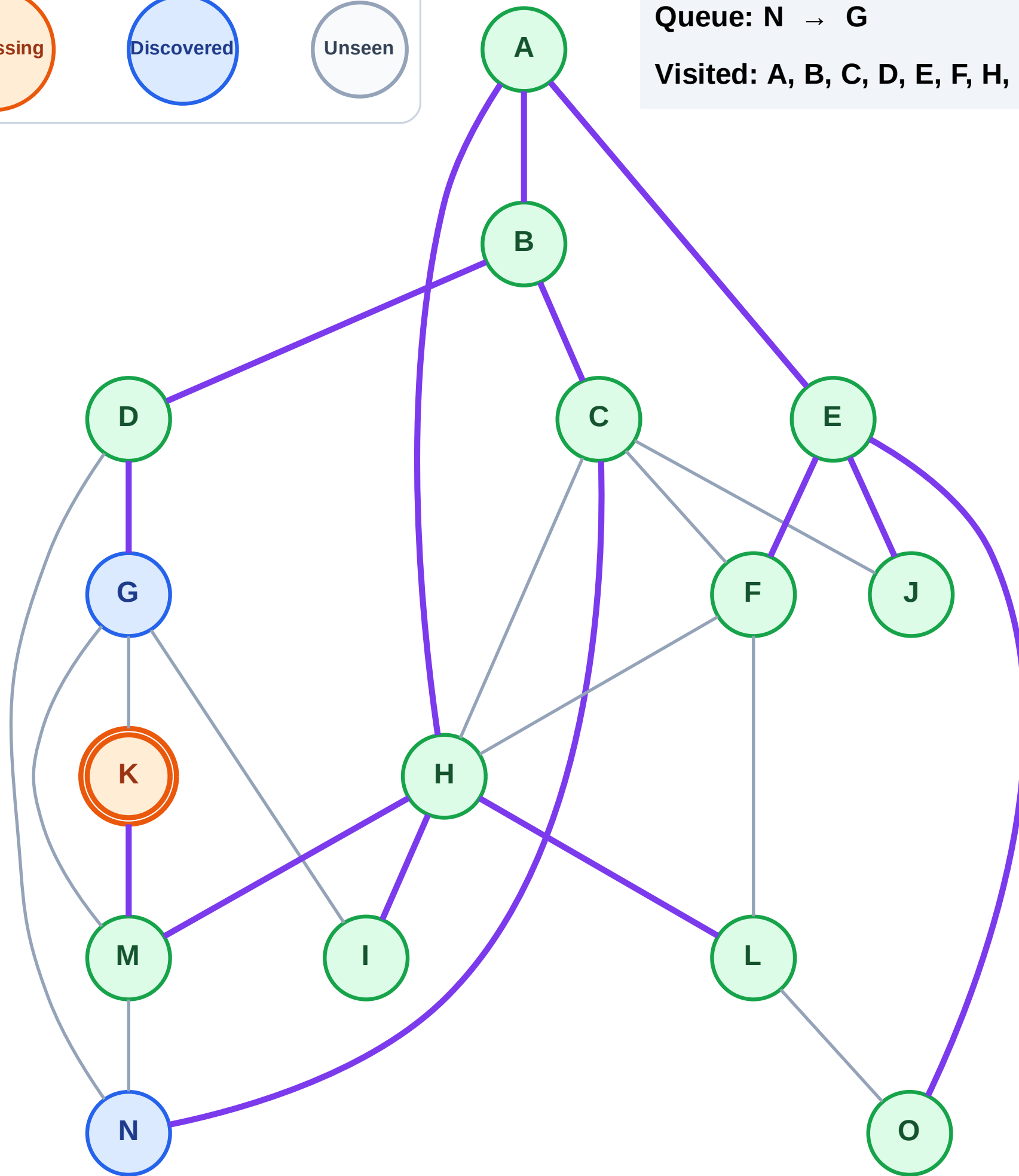
Step 25: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → H → M → K

Legend



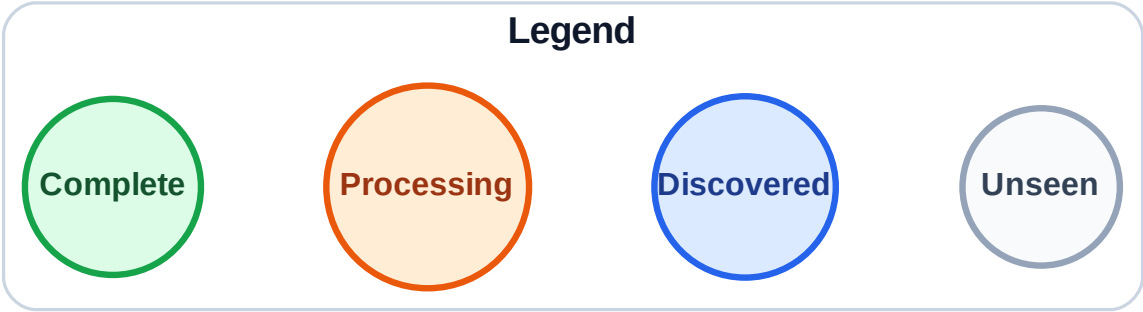
Queue: N → G

Visited: A, B, C, D, E, F, H, I, J, K, L, M, O

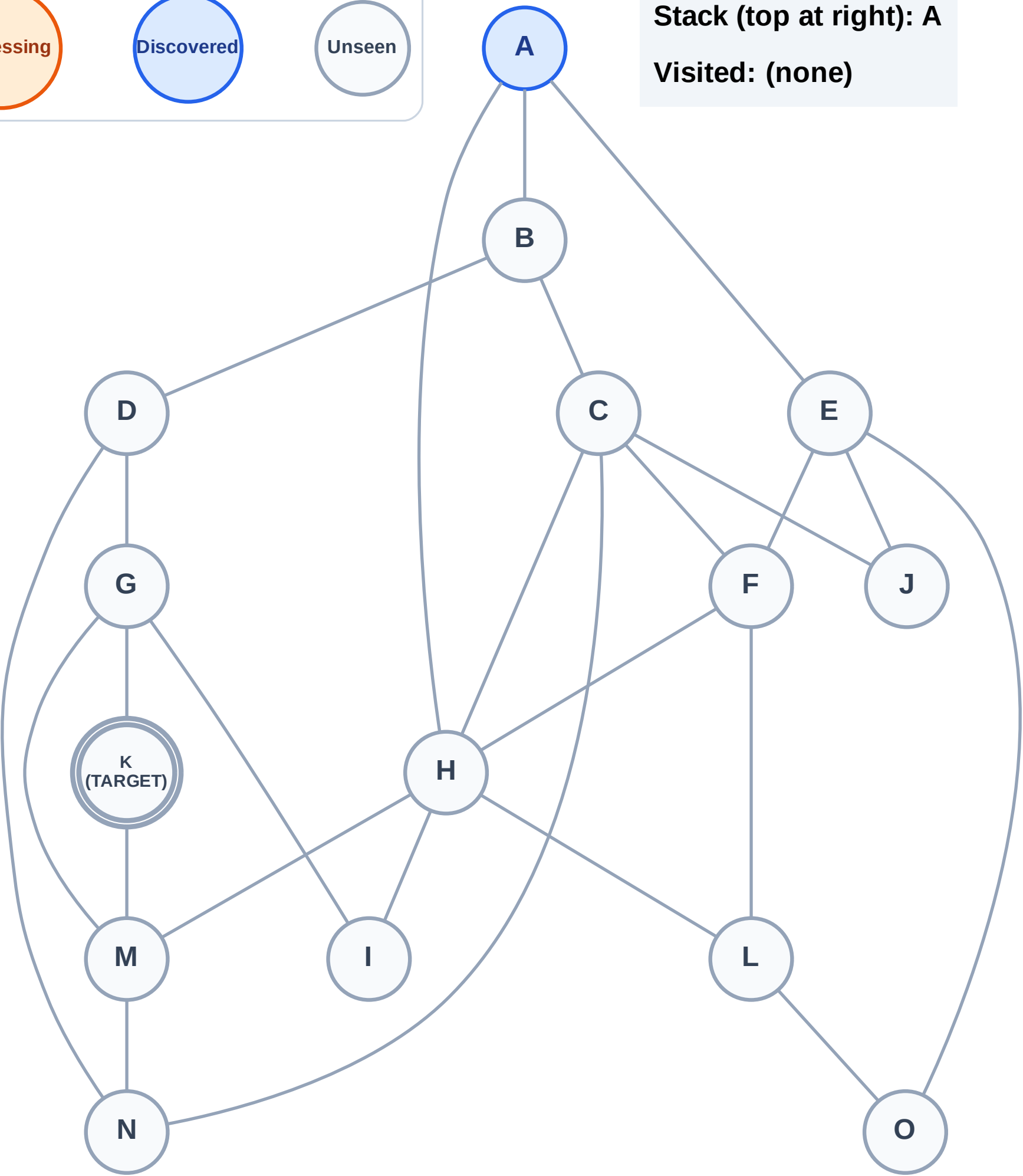


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

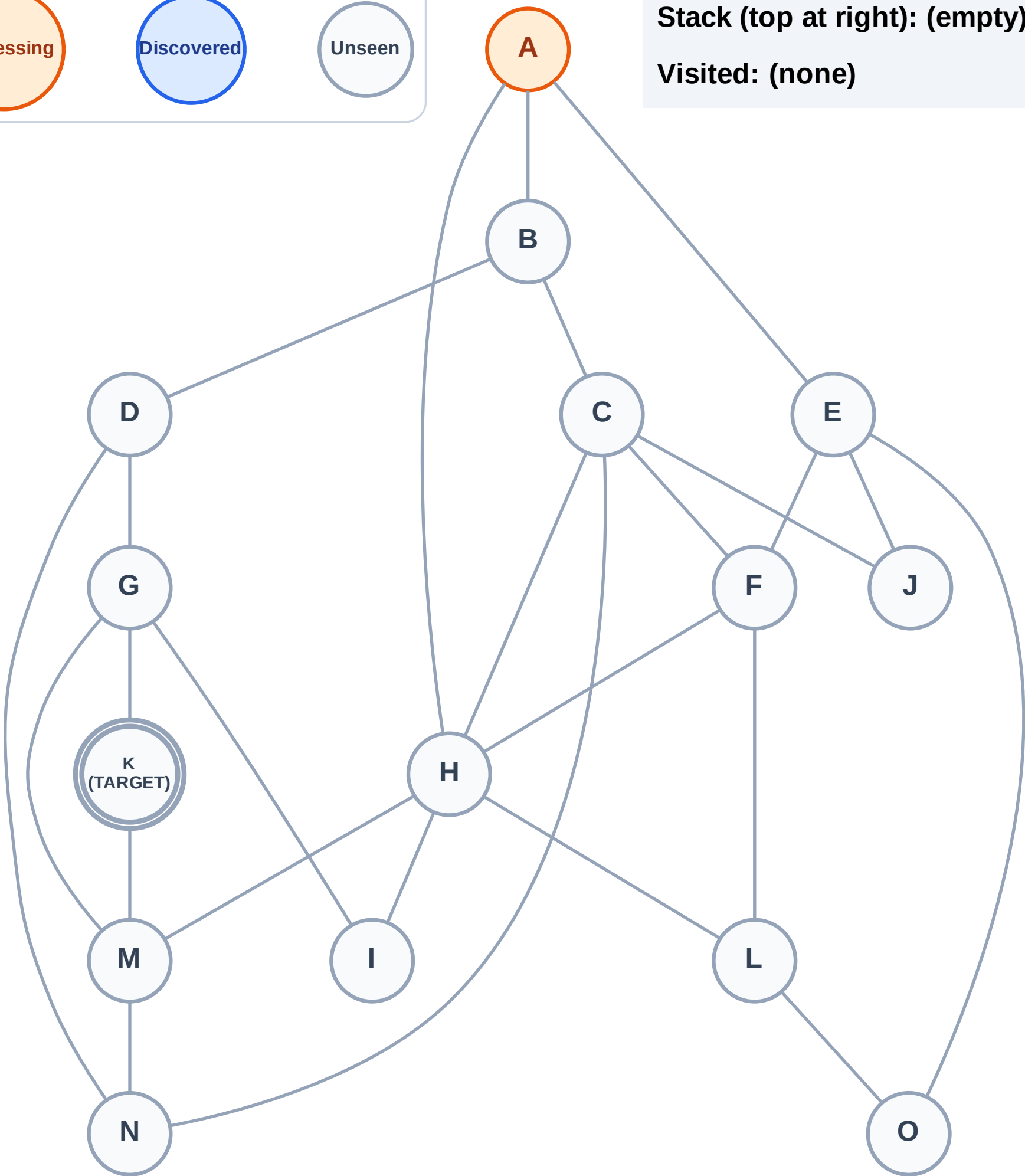
Complete

Processing

Discovered

Unseen

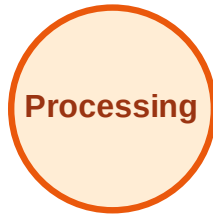
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

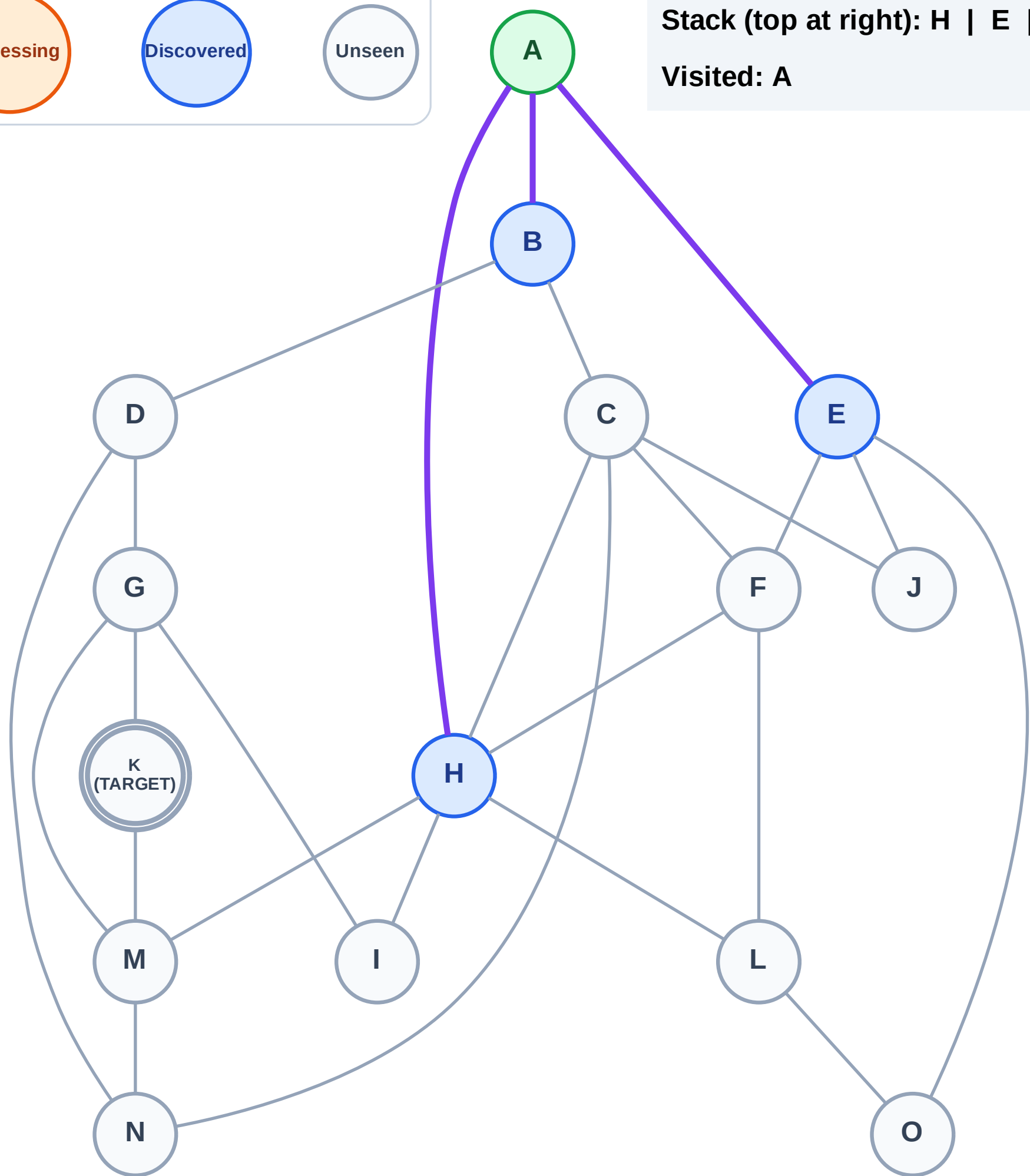
Step 3: Discover H, E, B from A

Legend



Stack (top at right): H | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

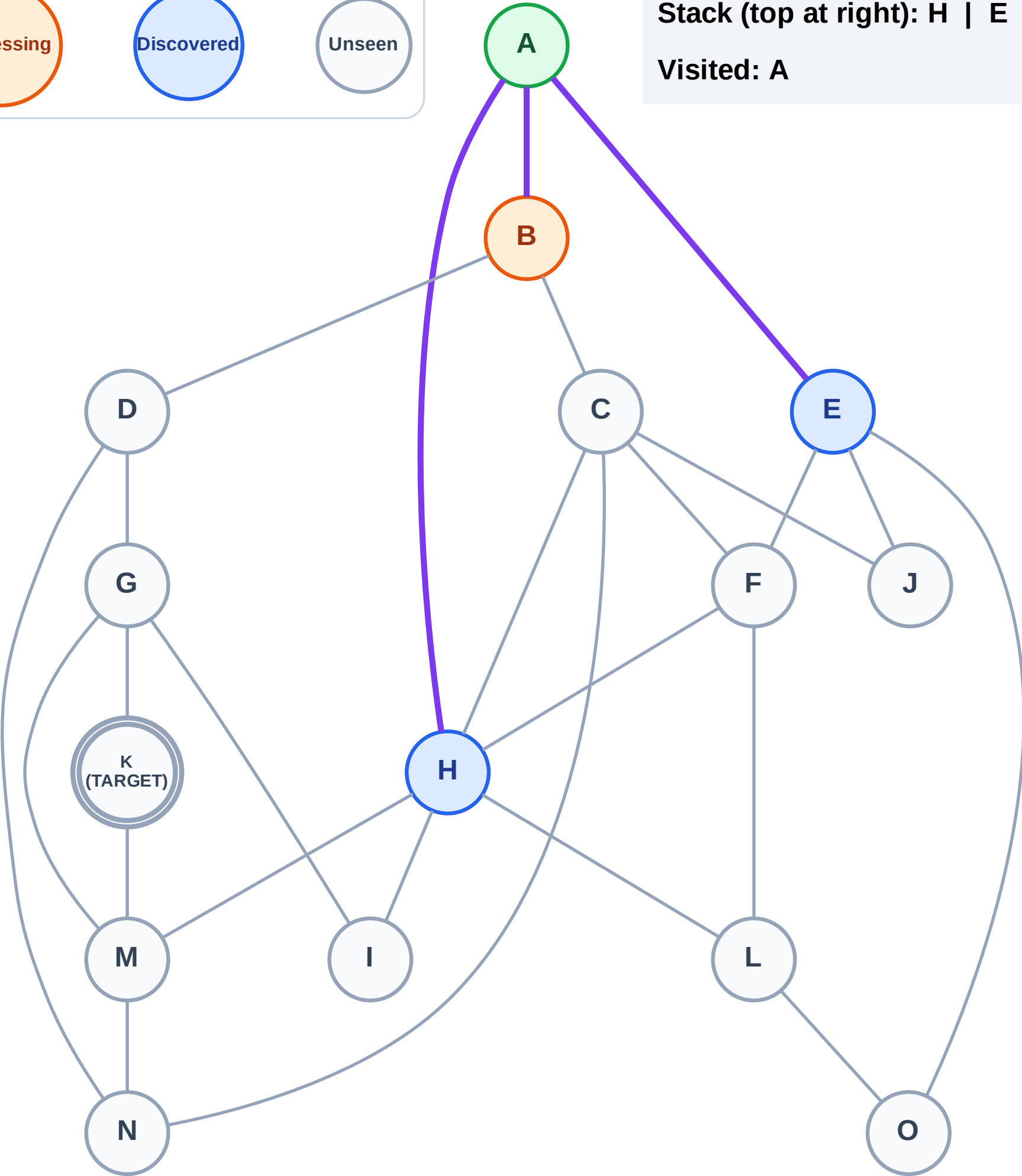
Complete

Processing

Discovered

Unseen

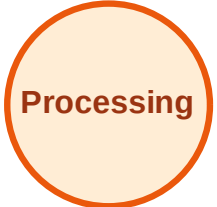
Stack (top at right): H | E
Visited: A



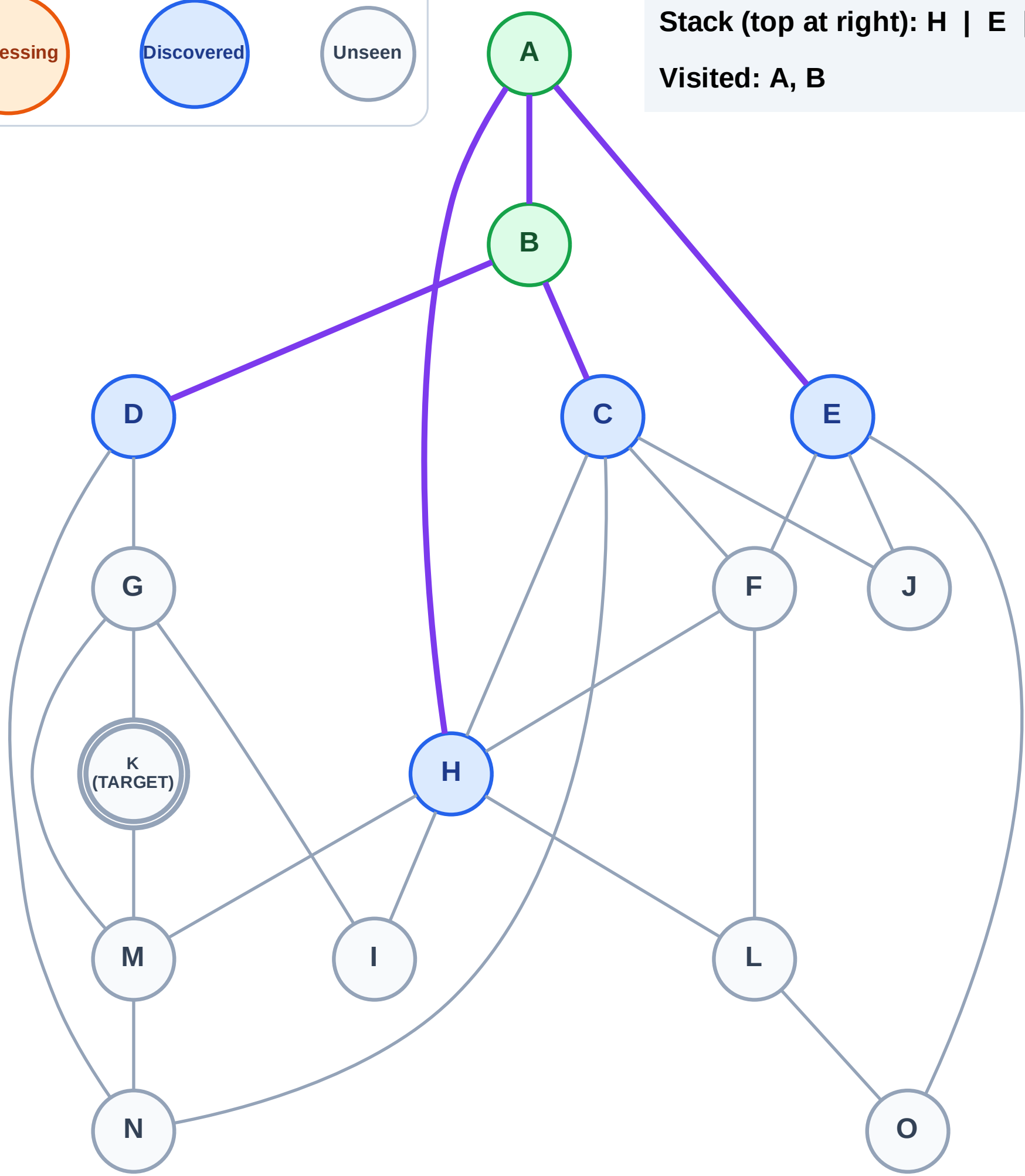
DFS Traversal

Step 5: Discover D, C from B

Legend



Stack (top at right): H | E | D | C
Visited: A, B



DFS Traversal

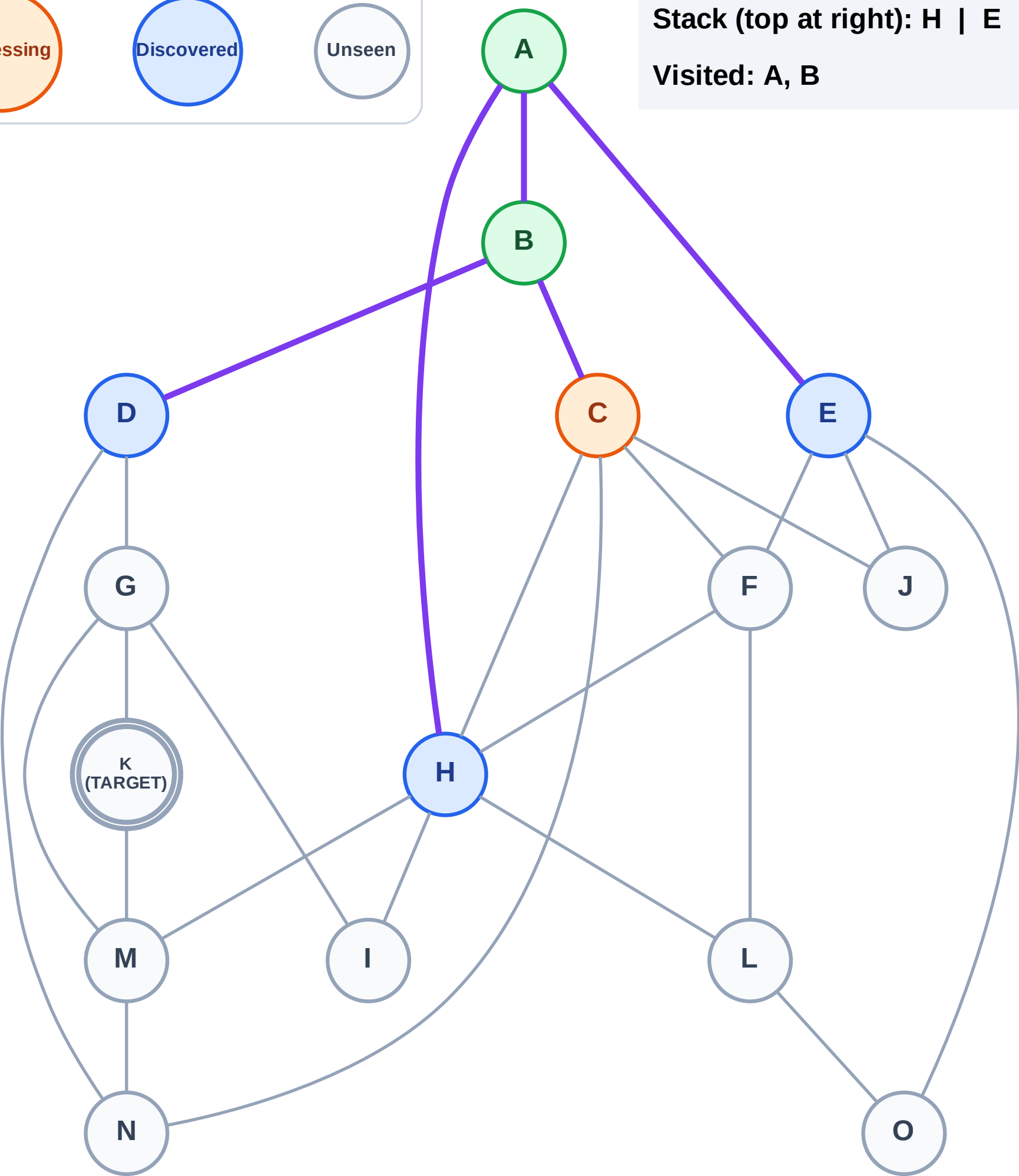
Step 6: Process node C

Legend



Stack (top at right): H | E | D

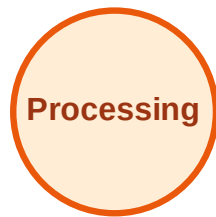
Visited: A, B



DFS Traversal

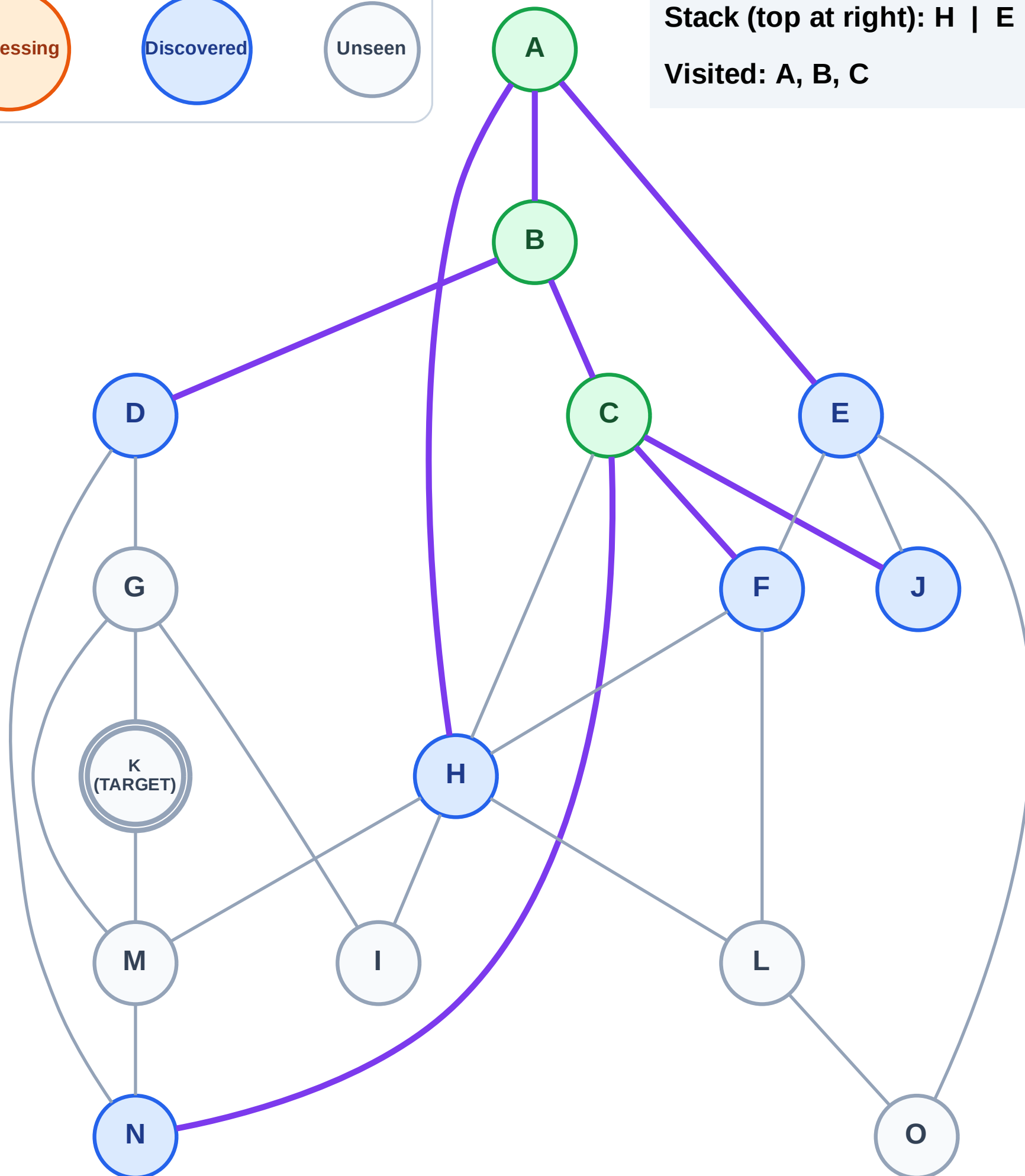
Step 7: Discover N, J, F from C

Legend



Stack (top at right): H | E | D | N | J | F

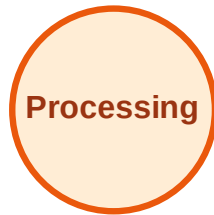
Visited: A, B, C



DFS Traversal

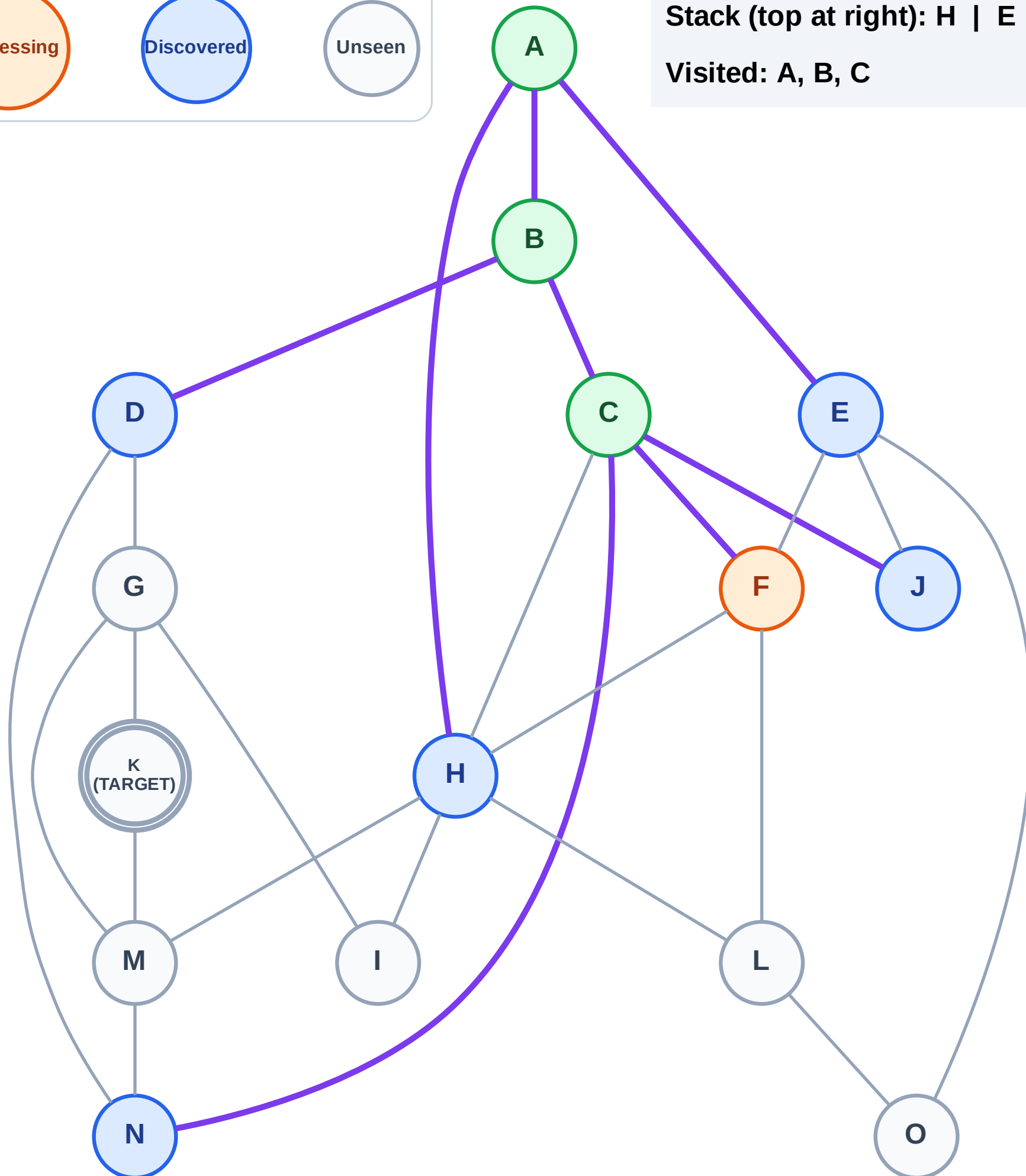
Step 8: Process node F

Legend



Stack (top at right): H | E | D | N | J

Visited: A, B, C



DFS Traversal

Step 9: Discover L from F

Legend

Complete

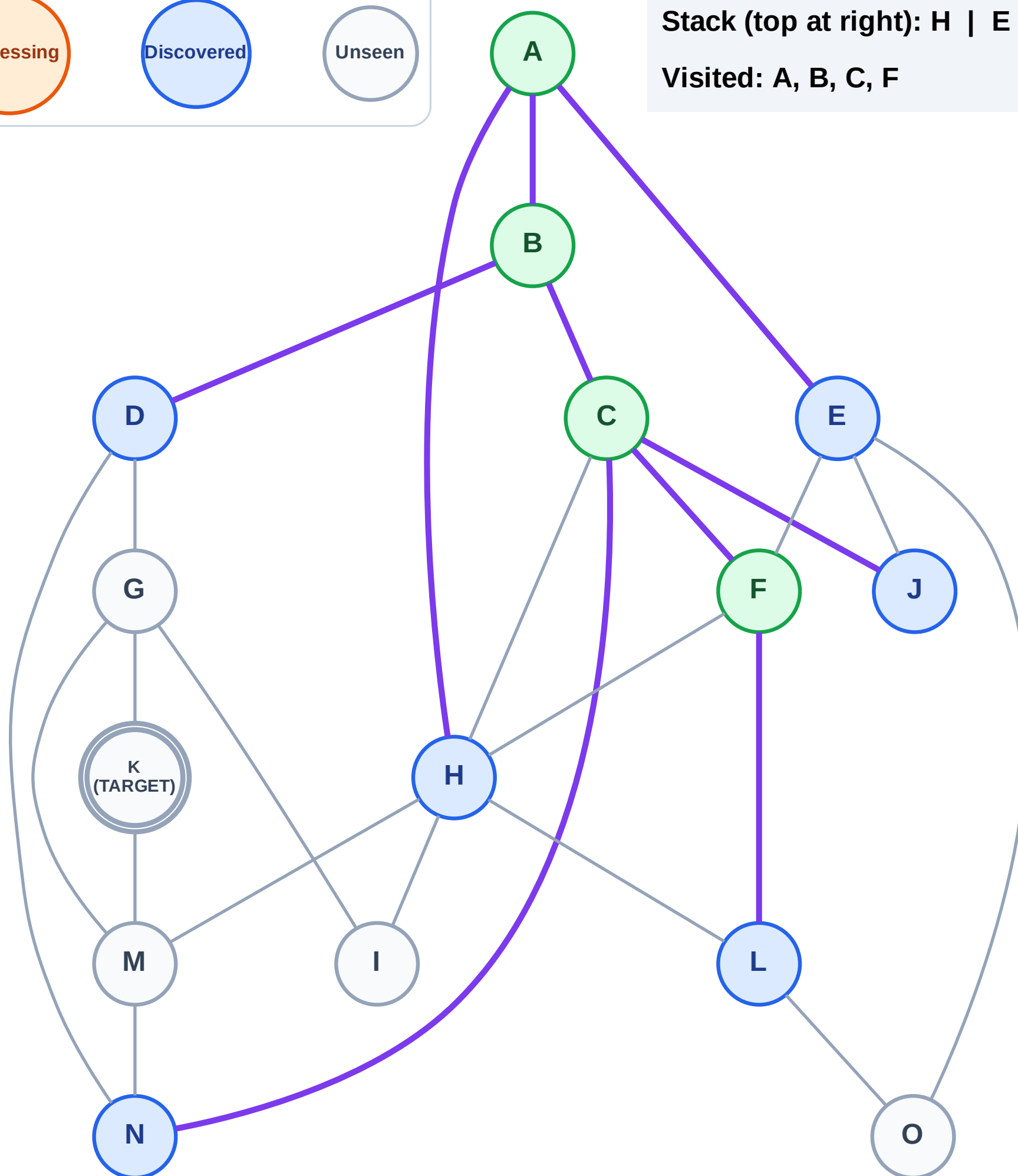
Processing

Discovered

Unseen

Stack (top at right): H | E | D | N | J | L

Visited: A, B, C, F



DFS Traversal

Step 10: Process node L

Stack (top at right): H | E | D | N | J
Visited: A, B, C, F

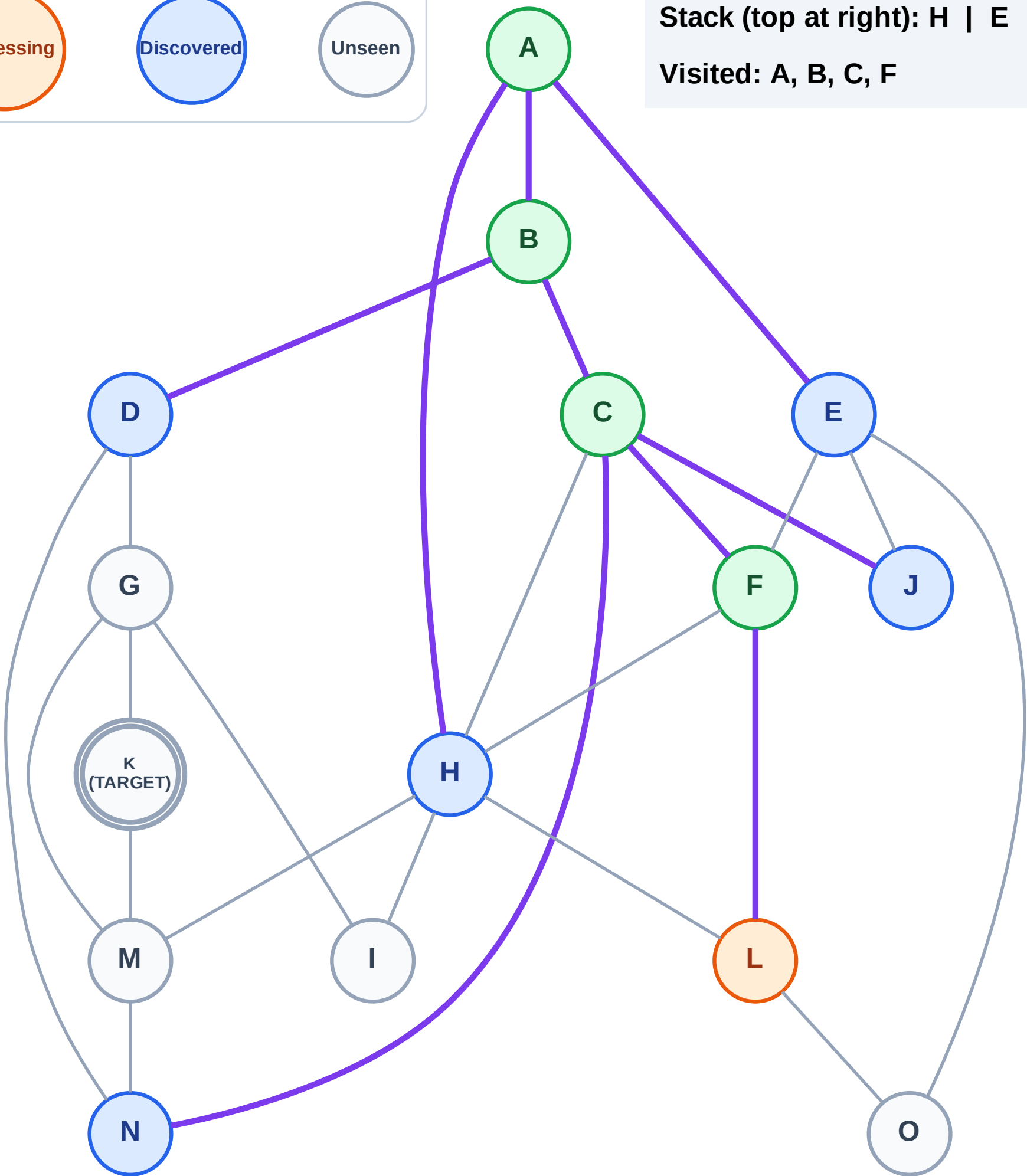
Legend

Complete

Processing

Discovered

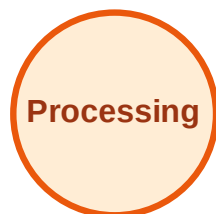
Unseen



DFS Traversal

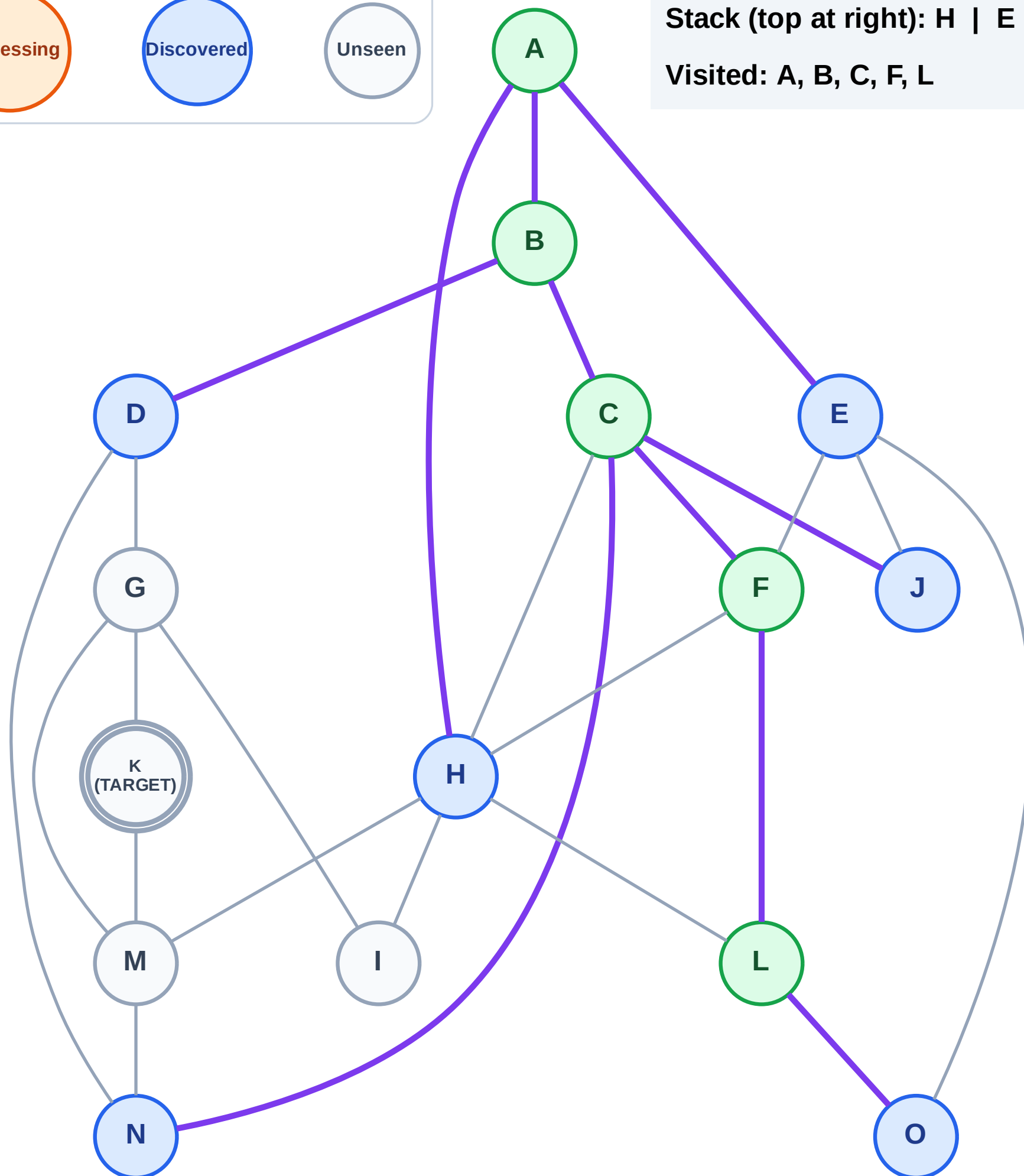
Step 11: Discover O from L

Legend



Stack (top at right): H | E | D | N | J | O

Visited: A, B, C, F, L



DFS Traversal

Step 12: Process node O

Legend

Complete

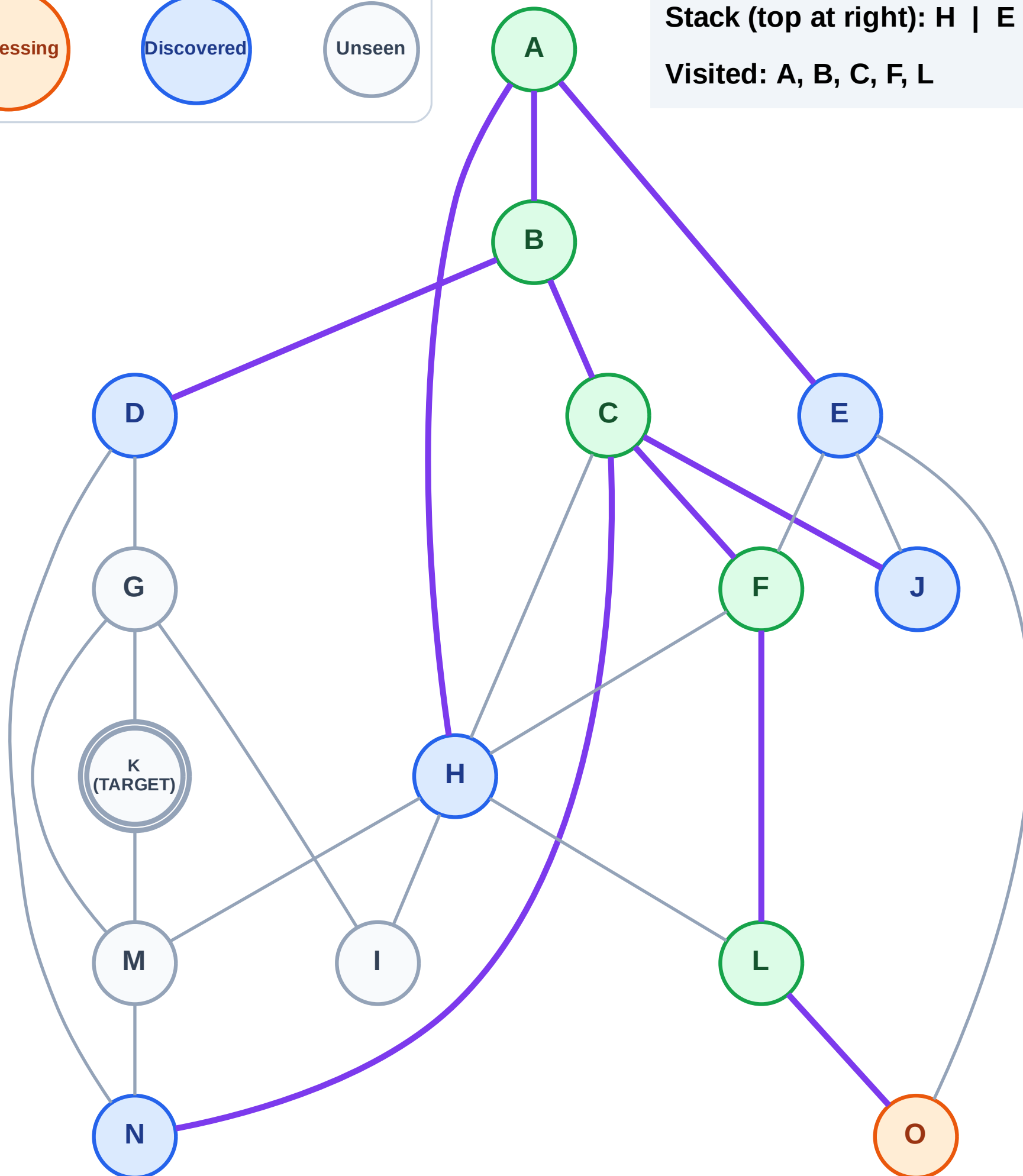
Processing

Discovered

Unseen

Stack (top at right): H | E | D | N | J

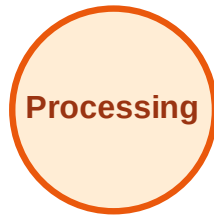
Visited: A, B, C, F, L



DFS Traversal

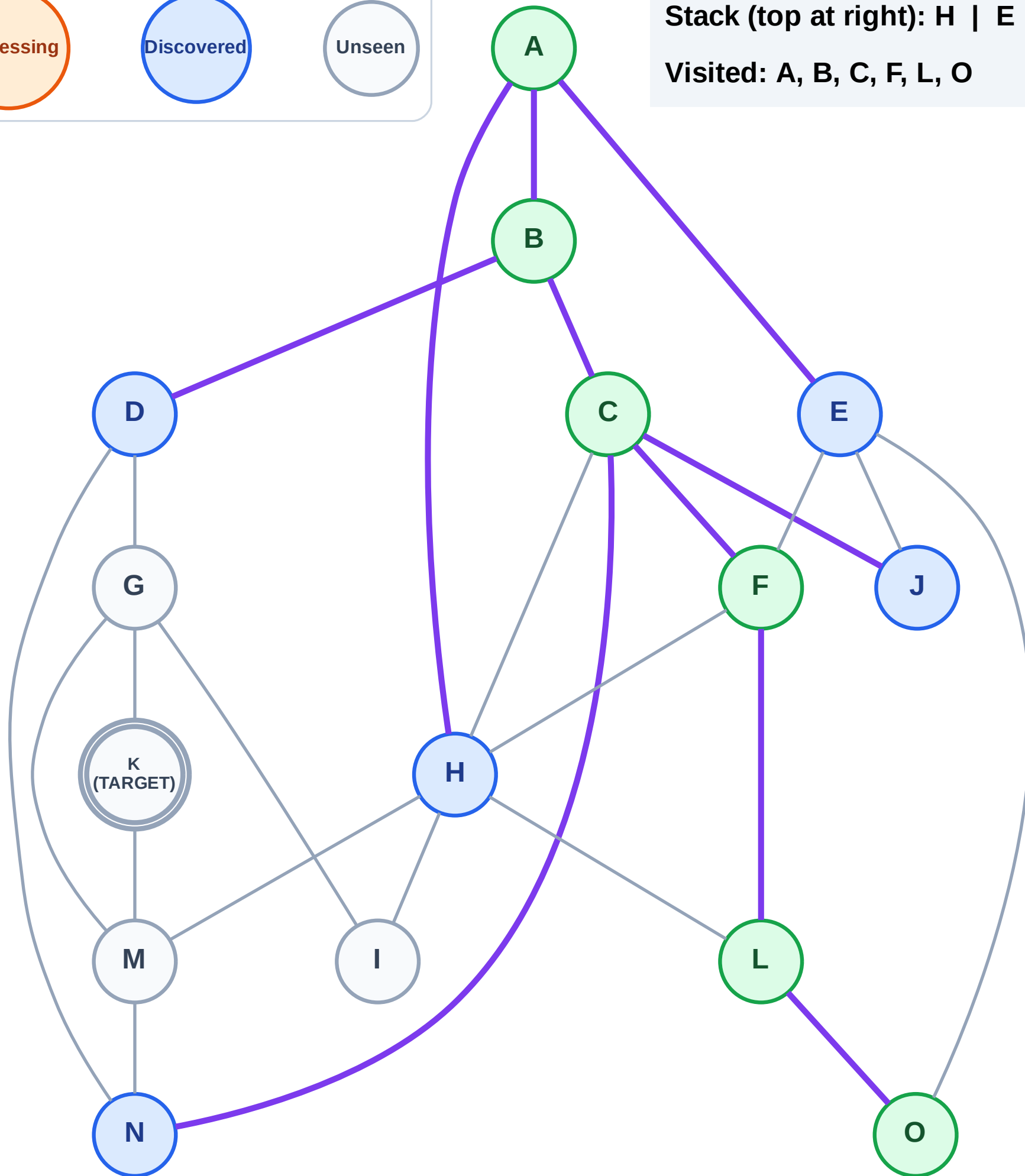
Step 13: No new neighbors from O

Legend



Stack (top at right): H | E | D | N | J

Visited: A, B, C, F, L, O



DFS Traversal

Step 14: Process node J

Legend

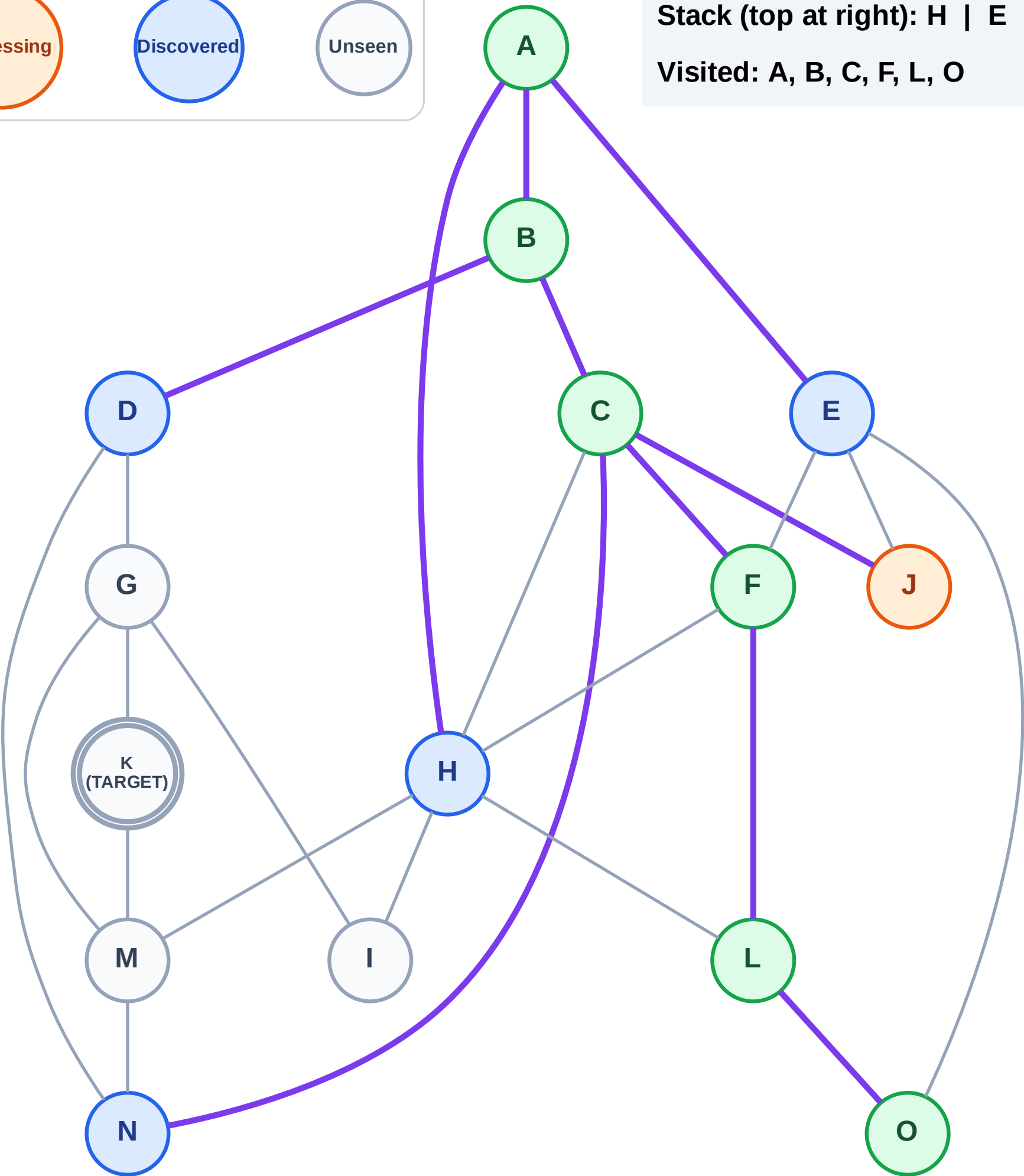
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | D | N
Visited: A, B, C, F, L, O



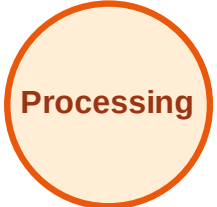
DFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



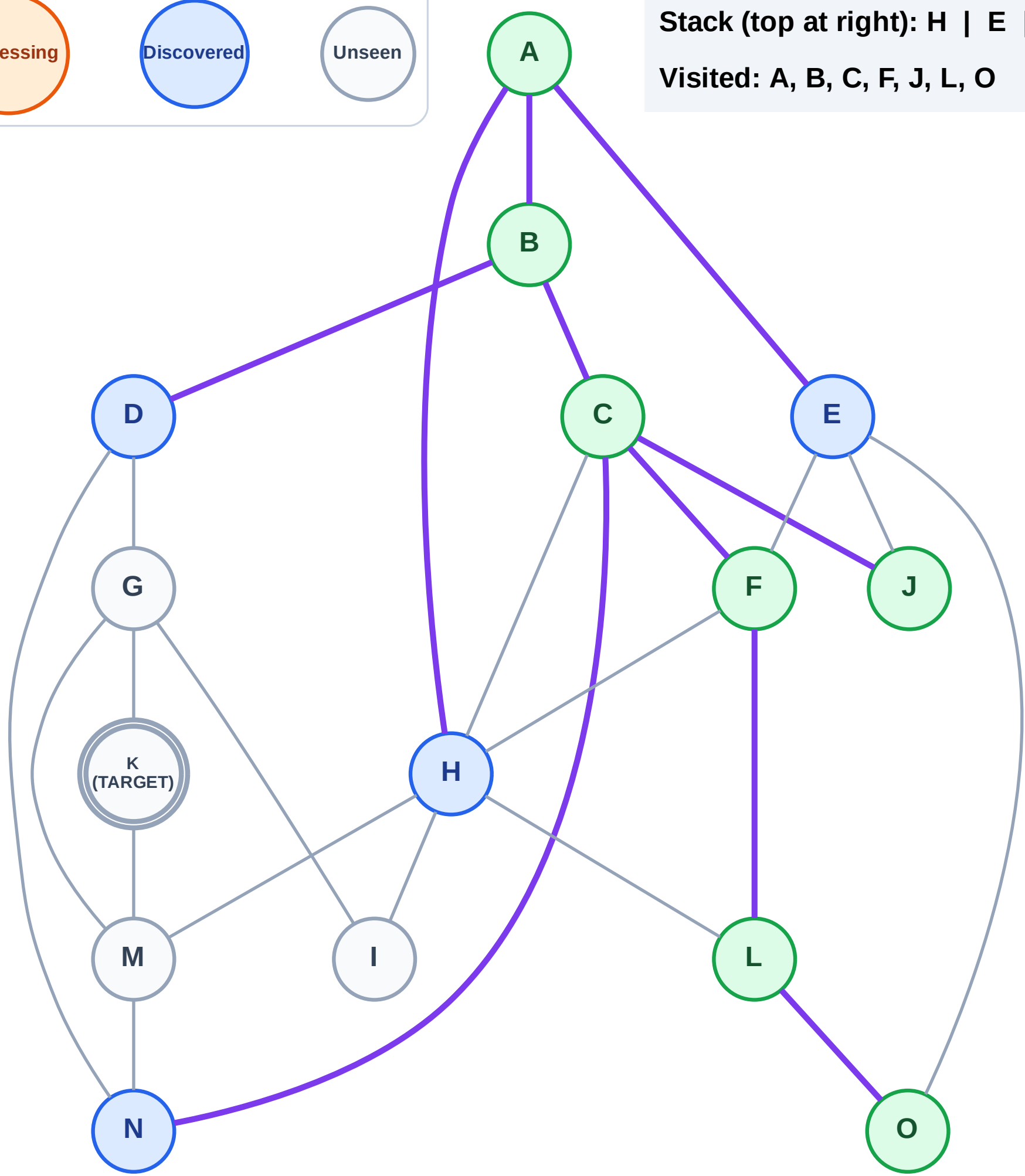
Discovered



Unseen

Stack (top at right): H | E | D | N

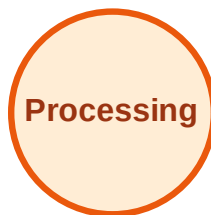
Visited: A, B, C, F, J, L, O



DFS Traversal

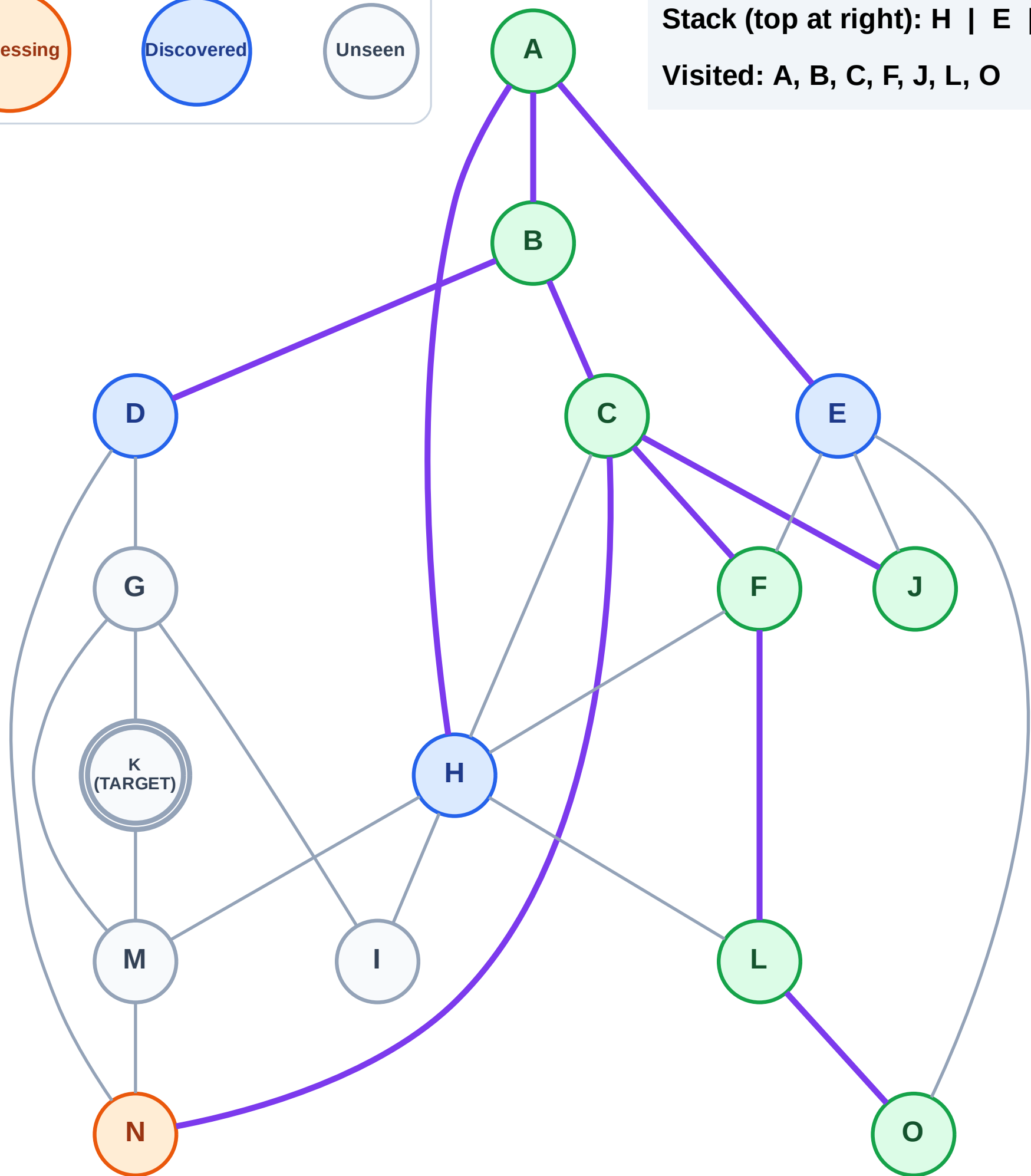
Step 16: Process node N

Legend



Stack (top at right): H | E | D

Visited: A, B, C, F, J, L, O



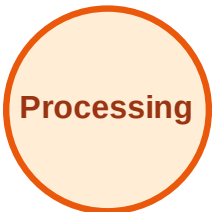
DFS Traversal

Step 17: Discover M from N

Legend



Complete



Processing



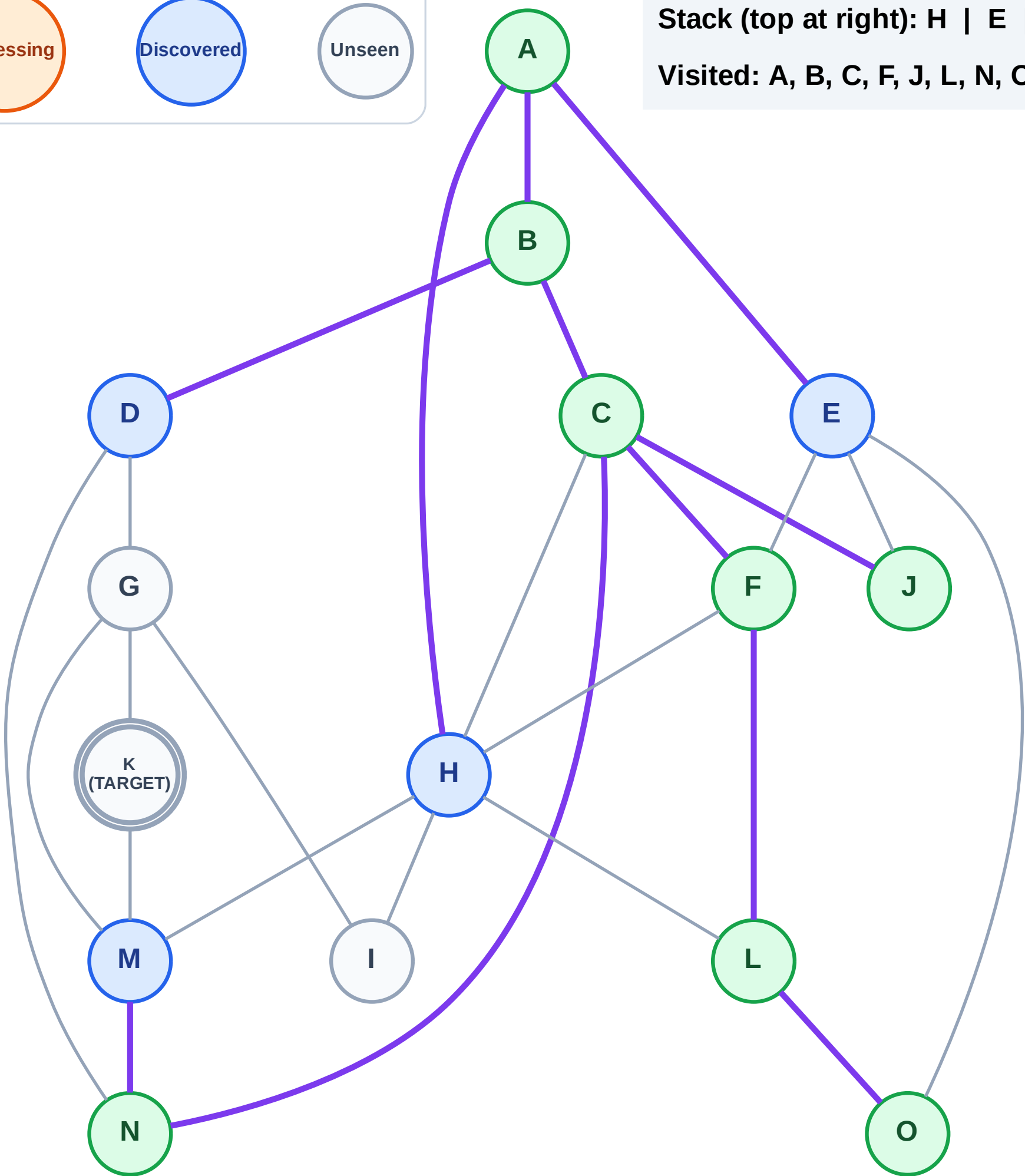
Discovered



Unseen

Stack (top at right): H | E | D | M

Visited: A, B, C, F, J, L, N, O



DFS Traversal

Step 18: Process node M

Legend

Complete

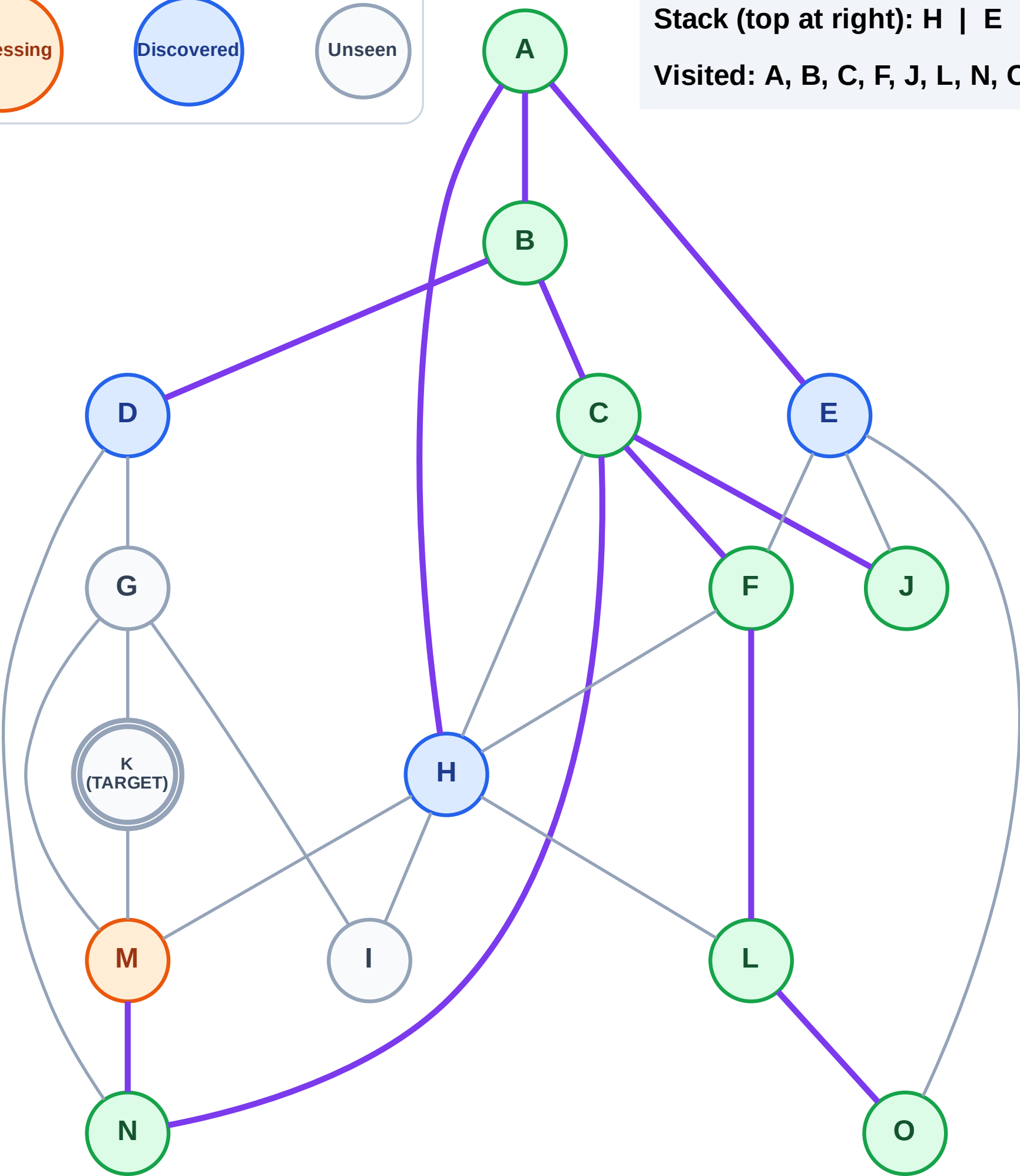
Processing

Discovered

Unseen

Stack (top at right): H | E | D

Visited: A, B, C, F, J, L, N, O



DFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → C → N → M → K

Legend



Stack (top at right): H | E | D

Visited: A, B, C, F, J, K, L, M, N, O

